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Diplomarbeit

Efficient Adaptive Global Illumination Algorithms

Universität Ulm
Fakultät für Informatik
Institut für Medieninformatik
Manuel Jan Kugelman, 2006

Betreuer/Gutachter: Prof. Dr. Alexander Keller, Universität Ulm
Zweitgutachter: Prof. Dr. Heiko Neumann, Universität Ulm

Ich danke ganz herzlich jedem einzelnen,
der mich auf irgend eine Weise bei der Erstellung dieser Arbeit unterstützt hat.

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Chapter 1

Introduction

The realistic simulation of light transport is one of the very basic assets for the generation of photorealistic images from technical scene descriptions. Since the physically correct simulation of light transport is still not feasible for batch image generation, we have to resort to simplifications and coarse solutions of the *Global Illumination Problem*.

The measurements of incoming radiance necessary for image generation are typically estimated by numerical integration techniques. To extract a sufficient amount of information from a limited number of samples, it is necessary to concentrate on the most important information, and to cleverly reuse the obtained information. Insufficient information results in noisy images and visible artifacts. Therefore efficient sampling techniques that allow for adaptation to the characteristics of light transport are essential to generate high quality images for low costs.

This thesis presents various approaches to adaptive sampling for light transport problems. It introduces a simple technique to adapt the evaluation efforts for direct illumination by point light sources to the expected local variance of the illumination. A byproduct is the investigation of some simple illumination prediction methods. The simple initially introduced technique is generalized to cover the generic unbiased combination of approximations with corrective sampling. The generalized technique is used to design an efficient and unbiased adaptive sampling framework. Furthermore an unbiased variant of the *Light Cuts* algorithm is delivered. Another part of the thesis addresses the *Robust Instant Global Illumination* algorithm. Its range of application is extended to glossy scenes by introducing a composite bound for use with the robust bounding heuristic. Caustics are either handled by backtracing light subpaths or tailoring the sampling rate for light subpaths to the expected variability of the illumination. A technique to generate efficient point light sources by adapting progressively to the characteristics of light transport concludes the main part of the thesis. The section on future work gives hints to worthwhile future investigations.

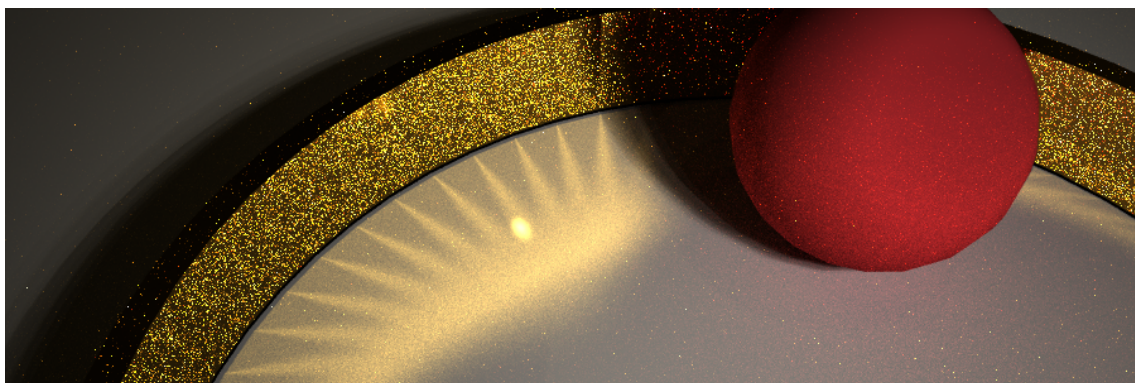


Figure 1.1: This image displays global illumination effects and a caustic, caused by a glossy faceted ring in conjunction with a small area light source. Different sampling rates were applied to sample low and high frequency illumination effects. A smooth low frequency lighting and a vivid directly seen caustic is achieved. However, the mirror image of the caustic is sampled insufficiently.

Annotations

In practice simulated light transport is typically based on geometric optics and assumed to be instantaneous and linear. While the effect of the limited speed of light is not observable to a notable extent in everyday settings, the bending of light paths is quite common due to continuously varying refraction indices as encountered in heated air. We furthermore limit the discussion in this work to scenes without participating media as for example fog or smoke. Provision for participating media is straightforward in principle, since a participating medium is nothing else than the sum of very many individual particles. Due to the extremely large number and small size of these particles the intersection and scattering are typically modeled through random processes that cause many cumbersome issues in practice. See the works of [Pauly et al., 2000], [Raab, 2005], and [Seibert, 2006] for a discussion of problems and solutions associated with participating media in a similar context.

To ensure a good readability we adjust the direction of imaginary light flow to a reading order from left to right throughout all equations on light transport. For clarity in equations of expected values and variance the probabilities of random selections are written in bold face. Thence the estimator corresponding to a specific structure of the equation is easy to deduce. Any occurrence of ξ denotes a uniform random variable on the interval $(0, 1]$.

We highly recommend the reading of the landmark thesis by [Veach, 1997] to acquire a fine grasp of simulated light transport.

Chapter 2

Global Illumination

A photo realistic image is the result of measuring the incoming radiance with a sensor array. Examples for such sensor arrays are the retina of the human eye, a CCD-chip in a digital camera or a photosensitive film in a traditional camera. Radiance $\frac{W}{m^2 sr}$ is radiant power per area and solid angle. Radiance is defined as the number of photons per time passing through a differential surface $dA_{\hat{\omega}}^{\perp}(\mathbf{x})$ perpendicular to a direction $\hat{\omega}$ with their directions contained in a differential solid angle $d\sigma(\hat{\omega})$ around $\hat{\omega}$. A radiance measurement I_j is given by the sensor response for the sum of all photons incident at the sensor. If we consider the density of photons to be continuous, the sensor response for the sum of all incident photons transforms to the sensor response for the radiance transport along all possible transport paths from a light source to a sensor. The natural mathematical description is the integration of the sensor response for the radiance transported along a single path over the space of all transport paths from a light source to a sensor. The appropriate mathematical framework for rendering algorithms has been developed by [Veach, 1997].

2.1 Path Integral Formulation of Light Transport in Vacuum

A transport path $\bar{x}_k$ with $k + 1$ vertices $(\mathbf{x}_0, \dots, \mathbf{x}_k)$ is the sequence of all points, at which a particle traveling along the path interacts with the scene boundaries.

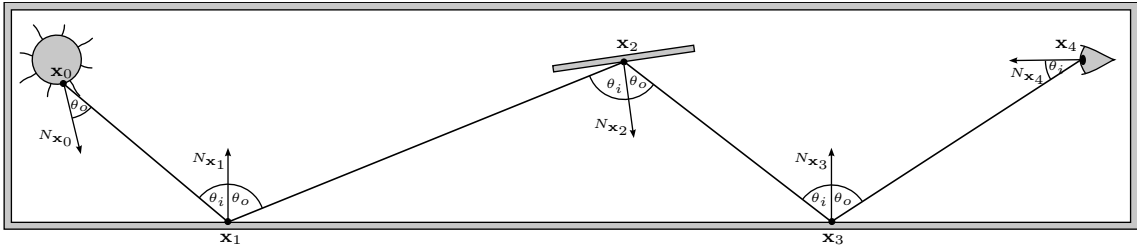


Figure 2.1: A complete transport path $\bar{x}_4 = (\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4)$ annotated with the normals $N_{\mathbf{x}_i}$ at its vertices $\bar{x}_i$ and their angles θ_i and θ_o to the incoming and outgoing edge.

Consider the set Ω_k of all paths $\bar{x}_k$ with length k . We define a product measure on its subsets $D_k \subset \Omega_k$. [Veach, 1997] chose the *area product measure* $\mu_k^A(D_k)$ for its simplicity. The *area product measure* is the product of the *area measures* from the regions where the transport paths may interact with the scene surface. In the case of point shaped interaction regions the differential area measure has to be used.

$$\mu_k^A(D_k) = \int_{D_k} dA(\mathbf{x}_0) \dots dA(\mathbf{x}_k) \quad (2.1)$$

Path space Ω is the set of transport paths of all possible lengths. We combine the sets of paths of all finite

lengths Ω_k to form the path space Ω with an *extended area product measure* $\mu^a(D)$.

$$\Omega = \bigcup_{k=0}^{\infty} \Omega_k \quad (2.2a)$$

$$\mu^a(D) = \sum_{k=1}^{\infty} \mu_k^A(D \cap \Omega_k) \quad (2.2b)$$

The intersection $D \cap \Omega_k$ decomposes the set D into the disjoint subsets D_k whereof each consists solely of paths of the length k . The measure μ^a of D is the sum of the individual measures μ_k^A of all subsets D_k .

By using the defined domain and measure we can write the measurement I_j as an integration over path space:

$$I_j = \int_{\Omega} f_j(\bar{x}) d\mu^a(\bar{x}) \quad (2.3)$$

The formulation as a simple integral allows to apply any general-purpose integration technique.

2.1.1 Measurement Contribution Function

The *measurement contribution function* f_j describes the contribution of the path $\bar{x}_k = (\mathbf{x}_0, \dots, \mathbf{x}_k)$ to the measurement I_j formulated by equation 2.3. The summands of the *extended area product measure* are non-zero solely for paths of the length they are defined for. This fact allows to separate the different path lengths and enables us to define the integrand separately for every path length k :

$$f_j(\bar{x}_k) = L_e(\mathbf{x}_0 \rightarrow \mathbf{x}_1) \cdot \left(\prod_{i=1}^{k-1} G(\mathbf{x}_{i-1} \leftrightarrow \mathbf{x}_i) f_s(\mathbf{x}_{i-1} \rightarrow \mathbf{x}_i \rightarrow \mathbf{x}_{i+1}) \right) \cdot G(\mathbf{x}_{k-1} \leftrightarrow \mathbf{x}_k) W_{e,j}(\mathbf{x}_{k-1} \rightarrow \mathbf{x}_k). \quad (2.4)$$

Each path starts at a point $\mathbf{x}_0$ on a light source. The radiance emitted toward the point $\mathbf{x}_1$ is denoted by $L_e(\mathbf{x}_0 \rightarrow \mathbf{x}_1)$. An arbitrary number of scattering events at the intermediate interaction points $\mathbf{x}_1 \dots \mathbf{x}_{k-1}$ follows. The BSDF $f_s(\mathbf{x}_{i-1} \rightarrow \mathbf{x}_i \rightarrow \mathbf{x}_{i+1})$ describes the proportions of scattering from $\mathbf{x}_{i-1}$ to $\mathbf{x}_{i+1}$ at $\mathbf{x}_i$. The last point of a path lies on a sensor. The incoming radiance at the sensor is multiplied with the sensor responsivity $W_{e,j}(\mathbf{x}_{k-1} \rightarrow \mathbf{x}_k)$ for radiance arriving from point $\mathbf{x}_{k-1}$ at point $\mathbf{x}_k$. Each edge of the path introduces a *geometric term* $G(\mathbf{x}_{i-1} \leftrightarrow \mathbf{x}_i)$ that depends on the geometric relation between the two endpoints of the edge.

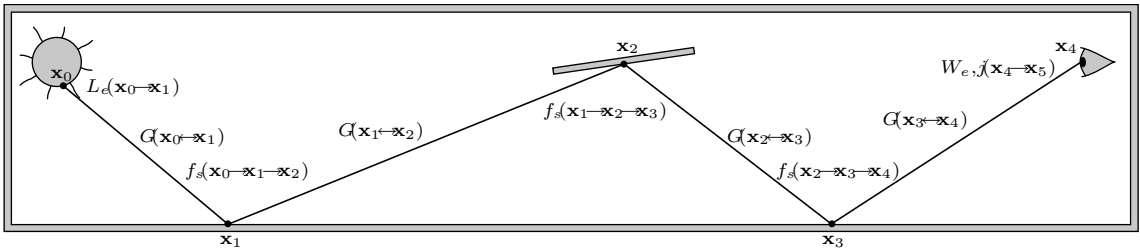


Figure 2.2: A complete transport path $\bar{x}_4 = (\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4)$. The edges and vertices are annotated with the corresponding terms of the measurement contribution function.

From an analytical point of view the geometric term is the Jacobian of the change from the differential projected solid angle measure to the differential area measure as given by $d\sigma_{\mathbf{x}_i}^\perp(\hat{\omega}_i) = d\sigma_{\mathbf{x}_i}^\perp(\widehat{\mathbf{x}_{i+1} - \mathbf{x}_i}) = G(\mathbf{x}_i \leftrightarrow \mathbf{x}_{i+1}) dA_{\mathbf{x}_i}(\mathbf{x}_{i+1})$. Geometrically it describes the differential projected area of the central projection of another differential projected area and equals the relative possible light transport between the two differential surface regions.

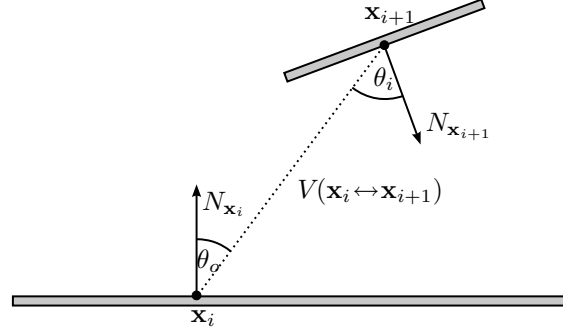


Figure 2.3: The Geometric Term

The geometric term is defined as

$$G(\mathbf{x}_i \leftrightarrow \mathbf{x}_{i+1}) = V(\mathbf{x}_i \leftrightarrow \mathbf{x}_{i+1}) \frac{|(\hat{\omega}_i \cdot N_{\mathbf{x}_i})(-\hat{\omega}_i \cdot N_{\mathbf{x}_{i+1}})|}{\|\mathbf{x}_i - \mathbf{x}_{i+1}\|^2} = V(\mathbf{x}_i \leftrightarrow \mathbf{x}_{i+1}) \frac{|\cos(\theta_{\mathbf{x}_{i_0}})\cos(\theta_{\mathbf{x}_{i+1_i}})|}{\|\mathbf{x}_i - \mathbf{x}_{i+1}\|^2}. \quad (2.5)$$

The *visibility function* $V(\mathbf{x}_i \leftrightarrow \mathbf{x}_{i+1})$ is 1, if the direct connection of $\mathbf{x}_i$ and $\mathbf{x}_{i+1}$ is possible, and 0 if it is blocked.

Insertion of equation 2.4 into equation 2.3 and explicit notation of the integration domain and measure reveal the full structure of the *path integral formulation*:

$$I_j = \sum_{k=1}^{\infty} \int_{\mathcal{M}^{k+1}} L_e(\mathbf{x}_0 \rightarrow \mathbf{x}_1) \cdot \left(\prod_{i=1}^{k-1} G(\mathbf{x}_{i-1} \leftrightarrow \mathbf{x}_i) f_s(\mathbf{x}_{i-1} \rightarrow \mathbf{x}_i \rightarrow \mathbf{x}_{i+1}) \right) \cdot G(\mathbf{x}_{k-1} \leftrightarrow \mathbf{x}_k) W_{e,j}(\mathbf{x}_{k-1} \rightarrow \mathbf{x}_k) dA(\mathbf{x}_0) \cdots dA(\mathbf{x}_k) \quad (2.6)$$

The measurement contributions for all individual path lengths k are summed. The measurement contribution from all paths of the length k is integrated over the space of the possible positions of their $k+1$ vertices. Without participating media this space typically is the space spanned by $k+1$ times the scene surface $\mathcal{M}$. The measure for an individual path is the product of the differential area measure of all vertices of the path.

2.2 Numerical Integration

With current methods and technology the analytic evaluation of the integral for a measurement is not feasible in general, except for simplistic cases. Therefore we have to evaluate the integral numerically.

An integration can be approximated by any suitable quadrature rule. Most quadrature rules place restrictions on the integrated function like e.g. a minimum number of continuous derivatives.

$$I = \int_D f(x) d\mu(x) \approx \hat{I}_N = \sum_{n=1}^N w_n f(x_n) \quad (2.7)$$

The weights w_i and sample points x_i are determined in advance by the quadrature rule. For N sample points typically a convergence rate of $O(N^r)$ with an integer $r \geq 1$ is achieved for an one-dimensional integral. It degrades to $O(N^{-\frac{r}{s}})$ for an s -dimensional integral solved by a tensor product rule. In fact *Bakhvalov's Theorem*, adverted by [Davis and Rabinowitz, 1984], generalizes this result to any s -dimensional quadrature rule for a function f with r continuous and bounded derivatives. This is called the *Curse of Dimensionality* by [Sobol', 1991].

2.2.1 Monte Carlo Integration

For the integration of functions on the infinite-dimensional path space we resort to *Monte Carlo integration*. We interpret the integral we wish to evaluate as an expected value of a continuous random function $f(x)$ and estimate it with N random samples x'_n chosen according to the probability density p :

$$\int_D f(x) d\mu(x) = \int_D \frac{\mathbf{P}(\mathbf{x})}{p(x)} f(x) d\mu(x) = \int_D \frac{1}{p(x')} f(x') dP(x') \approx \hat{I}_N = \frac{1}{N} \sum_{n=0}^{N-1} \frac{1}{p(x_n)} f(x_n) \quad (2.8)$$

The sample points are not defined in advance, as it is the case for quadrature rules, but random. This makes Monte Carlo integration independent of the dimension of the integral. The *Central Limit Theorem* ensures a convergence rate of $O(N^{-\frac{1}{2}})$ independent of the smoothness of the integrated function. The sole requirement is that the function is of bounded variance.

The error of an Monte Carlo estimator is almost certainly lower than tree times the standard deviation of f divided by the square root of the number of samples N :

$$P \left(\left| \int_D f(x) d\mu(x) - \frac{1}{N} \sum_{n=0}^{N-1} \frac{1}{p(x_n)} f(x_n) \right| \leq \frac{3\sigma(f)}{\sqrt{N}} \right) \approx 0.997 \quad (2.9)$$

When using Monte Carlo methods, it is necessary to examine the properties of the used estimators. The most important properties are bias and consistency. An unbiased estimator is defined by the requirement that the expected value of the estimator equals the real value of the estimated quantity. A consistent estimator is defined by the requirement that the estimate converges to the correct value for an infinite number of samples.

A lot of general information and references to further sources on the topic of Monte Carlo Integration in Computer Graphics are found in the thesis of [Veach, 1997]. Quasi-Monte Carlo Integration is treated extensively in the works of [Kuipers and Niederreiter, 1974], [Sobol', 1991], [Niederreiter, 1992], [Heinrich], [Owen, 1994], and [Keller, 1996, 2001].

2.2.2 Variance Reduction

The maximum number of samples is limited, so we have to allocate the samples where they are the most useful i.e. significantly reduce variance. On the other side it is sensible to reduce samples where they have little use. In order to miss no important information there has to be a non-zero probability to create any valid and contributing path. If this is not the case the *Problem of Insufficient Techniques* occurs and the resultant image is unavoidable biased because parts of the integrand are not sampled at all.

Because there is a vast body of work on variance reduction techniques, we only cover the few that have widespread use in *Computer Graphics*. Further information is found in the works of [Cochran, 1963] and [Sobol', 1991]. [Veach, 1997] provides additional references.

Stratification

It is advantageous to have samples in every region of the integrand and leave no large gaps between samples to minimize the chance of misestimating the contribution of a relatively large part. In fact it is most advantageous to place samples at the greatest possible distance to each other, which ensures that there are no gaps or clumps in the sample distribution. This property is called *low discrepancy*.

Stratification is the placement of at least one sample in every piece of a decomposition of the whole integration domain. It is not clear how to stratify the infinite-dimensional Path Space in a practical way since for deterministic sample placement the *Curse of Dimensionality* is effective. Since stratification is a very wide field of research, we do not take it into account for the remainder of this thesis and leave that for future work. A completely different approach to sampling without gaps would be the use of continuous information instead of point sampling, but this is also far out of the scope of this thesis.

It is possible to further increase the convergence rate of Monte Carlo integration by the use of stratified, correlated samples in the form of randomized low discrepancy point sets. The resulting integration method is known as *randomized quasi-Monte Carlo integration*. Integration techniques of this class converge with the rate $O(N^{-\frac{s+1}{2s}})$ for functions from L^2 according to [Keller, 1998] and [Kollig, 2004].

The randomized quasi-Monte Carlo estimator $\hat{I}_{K,N}$ is the combination of pre-smoothing through N stratified, correlated samples with standard Monte Carlo integration using K realizations:

$$\hat{I}_{K,N} = \frac{1}{K} \sum_{k=1}^K \frac{1}{N} \sum_{n=1}^N f(x_{k,n}) \quad (2.10)$$

where

$X = \{x_1, \dots, x_N\}$ is a deterministic point set with low discrepancy.

$X_k = \{x_{k,1}, \dots, x_{k,N}\}$ is a randomized version of X .

the set of $\{x_{k,n} : k = 1, \dots, K\} \sim U[0, 1]^s$ is uniformly distributed for a fixed $n \in \{1, \dots, N\}$.

the sets $X_1, \dots, X_K$ are of low discrepancy.

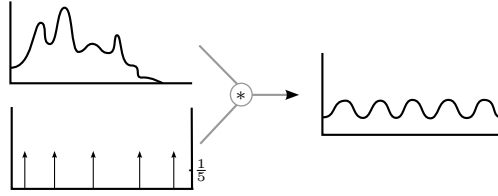


Figure 2.4: Schematic depiction of the pre-smoothing by five stratified point samples. The boundaries are assumed to be periodic.

Importance Sampling

An analytically integrable function g which is similar to the integrated function f can be used to guide the placement of samples through sampling according to a probability density function proportional to g :

$$\frac{f(x)}{g(x)} \approx \frac{1}{c} \text{ for } g(x) \approx cf(x)$$

$$E\left(\frac{f(x')}{g(x')}\right) = \int_{\Omega} \mathbf{g}(\mathbf{x}) \frac{f(x)}{g(x)} d\mu(x) = \int_{\Omega} f(x) d\mu(x). \quad (2.11)$$

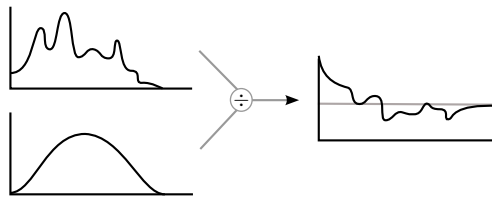


Figure 2.5: Importance Sampling does not suffer from oversampling a region but from undersampling. Therefore the critical point of the example setting is the severe undersampling at the left side.

It is not always possible to efficiently generate samples with the desired density or to normalize g to extract a probability density. This problems restrict the space of available probability density functions. Ideally the integrated function and the probability density used for *importance sampling* cancel out up to a constant of proportionality. The result would be an estimator with zero variance.

If the integrated function is a product of functions, we can use the knowledge on one of the factors to assess where the variance of the other factors is amplified the most and sample this regions more densely

by using the known factor as the probability density of the samples. The other unknown factors may still cause increased variance, if the factors unfortunately exhibit high variance in sparsely sampled regions. A common case of with this setup is the sampling of incoming radiance using a probability density according to the BRDF. We can capture the incoming radiance very well for smooth lighting conditions, but may increase variance for erratic lighting uncorrelated with the sampling density. This problem can be reduced by the supplemental use of other properties like approximated visibility or approximated incident light to determine the sample density. Spatial sampling of the light sources is also a kind of Importance Sampling for that the surfaces are sampled according to their emittance.

Most sampling techniques exhibit highly different performance for different parts of the integrand. If we have multiple densities for importance sampling that mimic different factors of the integrand, we would like to combine the results. A simple strategy to combine different techniques is to multiply the result of each strategy with a fixed weight and sum all results. The weights only have to sum up to 1 wherever the sampled function is non-zero. Another simple case are strategies which sample disjoint parts of the integrand. Their results are just added together.

Multiple importance sampling by [Veach and Guibas, 1995] is a general framework for the combination of different sampling strategies. The *multi sample estimator*, resembling a weighted sum of K techniques is given by

$$\hat{I} = \sum_{k=1}^K \frac{1}{N_k} \sum_{n=1}^{N_k} w_k(x_{k,n}) \frac{1}{p(x_{k,n})} f(x_{k,n}) \quad (2.12)$$

where $w_k(x)$ is the weighting function for the technique k .

The weighting functions which assign an individual weight to each combination of sample and technique, have to meet two conditions to preserve the unbiasedness of the estimate:

$f(x) \neq 0 \Rightarrow \sum_{k=1}^K w_k(x) = 1$ and $p_k(x) = 0 \Rightarrow w_k(x) = 0$. A rigorous proof is found in the thesis of [Veach, 1997].

[Veach and Guibas, 1995] developed a class of combination strategies which are provably good. The basic *balance heuristic* has weighting functions of the form

$$\hat{w}_k(x) = \frac{N_k p_k(x)}{\sum_l N_l p_l(x)}. \quad (2.13)$$

When inserted into the *Multi Sample Estimator*, this results in

$$\hat{I} = \frac{1}{N} \sum_{k=1}^K \sum_{n=1}^{N_k} \frac{1}{\sum_l c_l p_l(x_{k,n})} f(x_{k,n}) = \frac{1}{N} \sum_{k=1}^K \sum_{n=1}^{N_k} \frac{1}{\hat{p}(x_{k,n})} f(x_{k,n}) \quad (2.14)$$

where

$N = \sum_l N_l$ is the total number of samples.

$c_l = N_l/N$ is the fraction of samples from p_l .

$\hat{p}(x) = \sum_l c_l p_l(x_{k,n})$ is the combined sample density.

The combined sample density $\hat{p}(x)$ is the probability density for sampling the point x using the given allocation of samples to the different techniques. Weighting with the combined sample density ensures that the contribution of each sample is independent of the technique used to generate it. Although the Balance Heuristic is close to optimal, other weighting strategies work even better for special cases while retaining a similar worst case performance. If a sampling technique has a high relative probability of creating samples from a region, samples from this region get a weight close to 1. If the sample distribution from this technique matches the integrand very well in that region it would be beneficial to further increase the weights. If we assume that the sampling techniques are designed to match the integrand where they exhibit a high sample density relative to the other techniques and vice versa, it is beneficial to generally sharpen the weighting functions. Sharpened weighting functions reduce the contribution of samples that are generated with a low probability and increase the contribution of samples that are generated with a high probability. The cutoff heuristic and the power heuristic are examples of such strategies with sharpened weighting functions.

The *cutoff heuristic* can be written as

$$w_k(x) = \begin{cases} 0 & \text{if } N_k p_k(x) < \alpha \max_i (N_i p_i(x)) \\ \frac{N_k p_k(x)}{\sum_l \{N_l p_l(x) | N_l p_l(x) \geq \alpha \max_i (N_i p_i(x))\}} & \text{otherwise} \end{cases} \quad (2.15)$$

where a samples with a low weight are discarded.

The *power heuristic* is given by:

$$w_k(x) = \frac{(N_k p_k(x))^\beta}{\sum_l (N_l p_l(x))^\beta} \quad (2.16)$$

where the weights are raised to the power β and renormalized. For the choice of $\beta = 2$, the contribution of a sample is proportional to the relative probability density of its generation by the technique actually used to generate it.

These cutoff heuristic and the power heuristic share two limiting cases. In the case of $\alpha = 0$ and $\beta = 1$ the already introduced balance heuristic is obtained for $\alpha = 1$ and $\beta = \infty$ the maximum heuristic is obtained.

The *maximum heuristic* is given by:

$$w_k(x) = \begin{cases} 1 & \text{if } N_k p_k(x) = \max_i (N_i p_i(x)) \\ 0 & \text{otherwise} \end{cases} \quad (2.17)$$

where samples are only accepted if generated by the technique with the highest value for $N_k p_k(x)$.

A *robust bounding heuristic* can be obtained by reformulating the bounding of the geometric term for *Illumination in the Presence Of Weak Singularities* by [Kollig and Keller, 2004].

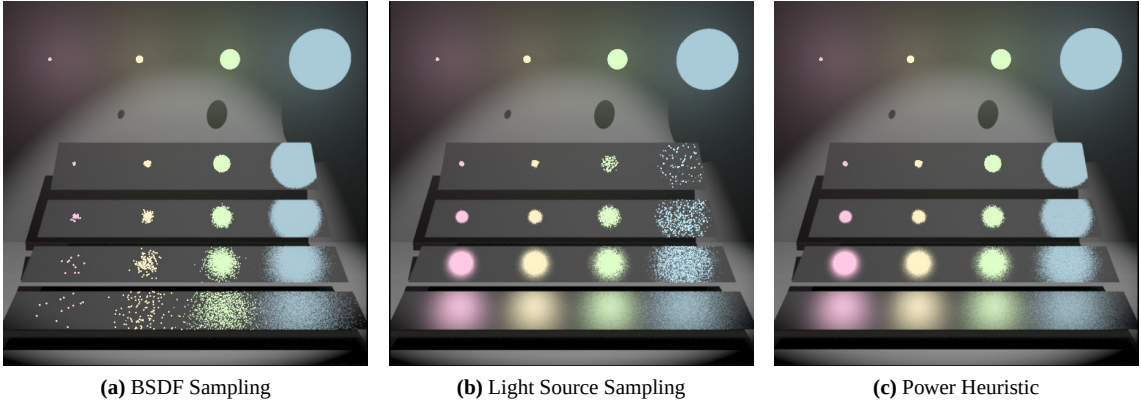


Figure 2.6: A example by [Veach, 1997] on the effect of multiple importance sampling. Directional importance sampling according to the BSGF and spatial sampling of the light source both fail for glossy highlights. The combination of both techniques using weights calculated by the power heuristic with $\beta = 2$ produces much better results.

Separation of the Main Part

If an approximation of the integrated function is available, we may use it to perform a *separation of the main part*. Therefor integrated function is separated into two additive parts: the approximated or analytically integrable main part and the unknown differences to the main part:

$$f(x) - g(x) \approx c \text{ for } g(x) \approx f(x) - c$$

$$E \left(\int_{\Omega} g(x) d\mu(x) + \frac{f(x)}{p(x)} \right) = \int_{\Omega} g(x) d\mu(x) + \int_{\Omega} \mathbf{p}(\mathbf{x}) \frac{f(x) - g(x)}{p(x)} d\mu(x) = \int_{\Omega} f(x) d\mu(x). \quad (2.18)$$

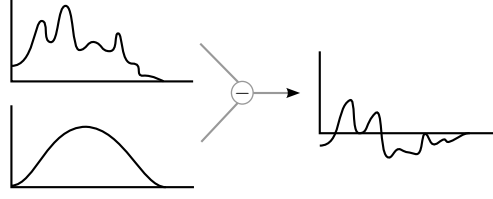


Figure 2.7: Separation of the Main Part solely depends on the difference between the main part and the function, which is largest at the spikes and valleys of the example function.

Another term for the *main part* is *control variate*. Ideally the difference between the main part and the real function is constant and thus an estimator with zero variance is available.

There is no benefit in using the same function as a control variate and density for importance sampling. As stated by [Kalos and Whitlock, 1986] using g as control variate is the better choice if $f - g$ is nearly constant. If $\frac{f}{g}$ is nearly constant then it is advised to use g for importance sampling. But importance sampling is also expedient in combination with a separation of the main part. The most significant information is not necessarily the knowledge of the highest function value and its location. If an approximation is available to be used as a control variate, the most significant information is the knowledge of the highest deviation from the approximation.

2.2.3 Efficiency

The efficiency of an estimator is defined as $\epsilon = \frac{1}{VT}$ where V is the variance of the estimator and T is the cost for evaluating the estimator. Therefore the efficiency may be increased not only by decreasing the variance but also by lowering the cost.

Extensive information reuse is a key to better images for less cost, because sensible reuse results in effectively more samples for the same cost. Of course the effect of correlated samples on the image quality has to be assessed. Information may be reused directly or as the input for interpolation and approximation. For reuse in any fashion the information has to be stored, selected, and if necessary the approximation or interpolation has to be calculated. Although reuse can save a lot of work, it introduces some overhead by itself. The challenge is to develop strategies for efficient reuse. This implies a sensible application of reuse i.e. preferable at low variance regions. Reuse in the presence of high variance is very likely wasted work and amplifies artifacts due to an insufficient sampling rate.

Splitting and Joining

If there are big differences in variance of the samples introduced by different dimensions of the integration domain, it is advisable to sample the more varying dimensions more densely. This can be achieved by *splitting* and is a simple example for partial path reuse. An exemplary estimator of a two-dimensional function $f(x, y)$ with splitting applied to the y -dimension is given by:

$$E \left(\frac{1}{p(x)} \frac{1}{K} \sum_{k=1}^K \frac{f(x, y)}{p_{|x}(y)} \right) = \int_{\Omega} \mathbf{p}(\mathbf{x}) \frac{1}{K} \sum_{k=1}^K \mathbf{p}_{|\mathbf{x}}(\mathbf{y}) \frac{f(x, y)}{p_{|x}(y)} d\mu(x) d\mu(y) = \int_{\Omega} f(x, y) d\mu(x) d\mu(y). \quad (2.19)$$

Joining is the opposite of splitting and effects the combination of multiple positions at one dimension with a single position at another dimension to form the complete position vectors of a group of samples. Both techniques, splitting and joining, lead to reuse of some dimensions of a sample location.

A typical use case for splitting is due to the differing spread of the scattering at different interaction points. A perfect mirror entails no spread and requires only a single sample to adequately capture the incoming radiance. A diffuse or moderately glossy surface requires much more samples to adequately capture non-constant incoming radiance.

The adaptation of the *Go with the Winners* strategy to light transport by citepgwtwkalos is an attempt to automate the splitting and joining of paths.

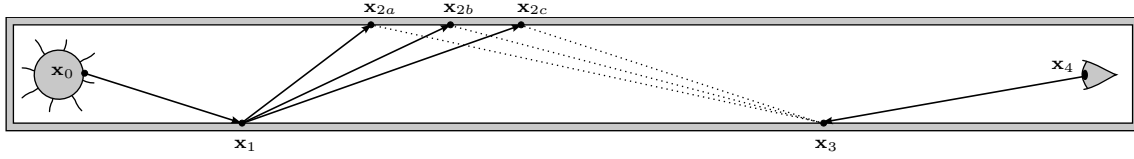


Figure 2.8: A path family created by splitting at x_1 and joining at x_3 .

Russian Roulette

Russian roulette, which was introduced to computer graphics by [Arvo and Kirk, 1990], increases the variance but lowers the cost. It is typically used where the expected additional variance is small compared to the reduced costs and thus the efficiency is increased. A common application is the truncation of paths to an average number of $\frac{1}{1-s}$ edges or the abortion of equation evaluation without causing bias. The original estimator $\tilde{e}(\cdot)$ is evaluated as usual with a probability of s and otherwise is omitted. To retain unbiasedness, the original estimator has to be weighted with $\frac{1}{s}$:

$$E \left(\begin{cases} \frac{1}{s} \tilde{e}(\cdot) & \text{if } \xi \leq s \\ 0 & \text{otherwise} \end{cases} \right) = s \frac{1}{s} E(\tilde{e}(\cdot)) + (1-s) \cdot 0 = E(\tilde{e}(\cdot)). \quad (2.20)$$

Russian roulette can also be interpreted as *Fractional Splitting* as done by [Szirmay-Kalos].

2.3 Algorithms

In this section we introduce some basic algorithms and techniques that allow for the sampling of path space by path tracing. We furthermore provide a glimpse of the most common rendering algorithms based on path tracing and their strengths and weak spots. We briefly assess to what extent the presented algorithms can adapt to the bidirectional characteristics of light transport and if reuse is employed.

2.3.1 Sampling Path Space

For numerical integration over path space we need to generate point samples from this infinite-dimensional space. A point sample from path space is a single path $\bar{x} = (\mathbf{x}_0, \dots, \mathbf{x}_k)$ consisting of a sequence of path vertices $\mathbf{x}_i$ corresponding to points of interaction between light and scene surfaces. The $k+1$ path vertices are connected by k path edges. The path integral form of light transport has no inherent limitations on the creation procedure of a transport path, but no general and robust sampling strategies to generate a complete path directly are known. Typically a path is generated step by step using local sampling techniques for the individual steps.

Recursive Path Generation

A practical way to create a complete light transport path is to trace the path recursively. After sampling the first vertex, the path is repeatedly extended by sampling the next interaction point. A edge between two points is only valid if the connection of the points is unobstructed. Typically Russian roulette is used to limit the number of edges of a path. At every step it is randomly decided if the path is terminated or continued. Another possibility to form a complete light transport path is to join multiple subpaths.

To compute the probability of a full path $\bar{x} = \mathbf{x}_0 \dots \mathbf{x}_k$ we just have to expand the density with respect to

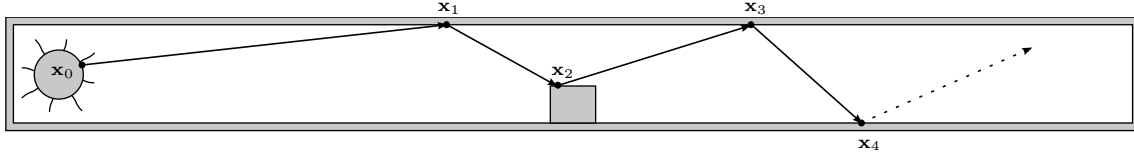


Figure 2.9: Recursive path generation: First x_0 is determined by sampling a position on the light source, then x_1 is acquired by directional sampling. The path is continued by directional sampling until Russian roulette indicates the termination of the path.

the *area product measure* to a product of densities in *area measure* for all interaction points:

$$p^{\mu^a}(\bar{x}_k) = \frac{dP}{d\mu^a}(\bar{x}_k) = \frac{dP}{d\mu^a}(\mathbf{x}_0 \cdots \mathbf{x}_k) = \prod_{i=0}^k \frac{dP}{dA}(\mathbf{x}_i). \quad (2.21)$$

Local Sampling Techniques

Local sampling techniques only use local information, i.e. information associated with the local interaction point. Examples are the position of the interaction point and the BSDF of the surface at the interaction point. For realistic scene configurations local sampling techniques suffice to sample all possible paths. When the scene configuration approaches singular settings, non-local techniques might improve the sampling quality. A treatise on the limitations of local sampling techniques is found in the thesis of [Veach, 1997].

The most common local sampling techniques are *directional sampling*, *spatial sampling*. To combine two points to an edge we have to do a *deterministic connection*.

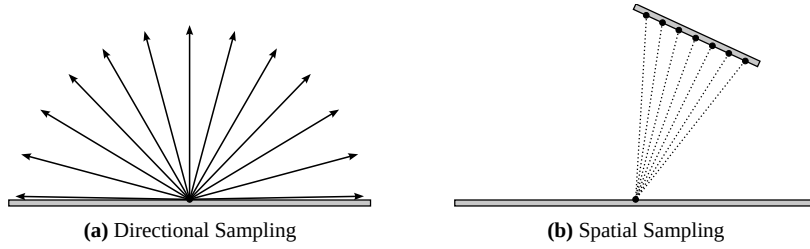


Figure 2.10: Sampling Techniques

To determine a point using *spatial sampling* we generate a point on the scene surface according to a given probability density. In the majority of cases we are interested in sampling points on the surfaces of the light sources according to the emitted radiance.

To create a point by *directional sampling* a starting point has to be determined. Then we sample a direction according to a given probability density, typically a density proportional to the BSDF of the surface at the starting point. To obtain the desired point we have to determine the point on the scene surface, that is closest to the starting point and is located at the sampled direction.

Since we use the area product measure to measure path space we have to express all sampling densities with respect to the area measure. The Jacobian of the change of variables from differential projected solid angle measure to the differential area measure is the geometric term as defined in equation (2.5). From this we calculate the conversion formula:

$$\frac{dP}{dA_{\mathbf{x}'}}(\mathbf{x}'') = \frac{dP}{d\sigma_{\mathbf{x}'}}(\widehat{\mathbf{x}'' - \mathbf{x}'}) \frac{d\sigma_{\mathbf{x}'}^{\perp}(\widehat{\mathbf{x}'' - \mathbf{x}'})}{dA_{\mathbf{x}'}(\mathbf{x}'')} = \frac{dP}{d\sigma_{\mathbf{x}'}}(\widehat{\mathbf{x}'' - \mathbf{x}'}) G(\mathbf{x}' \leftrightarrow \mathbf{x}'') \quad (2.22)$$

$$\Rightarrow p_{\mathbf{x}'}^A(\mathbf{x}'') = p_{\mathbf{x}'}^{\sigma^{\perp}}(\widehat{\mathbf{x}'' - \mathbf{x}'}) G(\mathbf{x}' \leftrightarrow \mathbf{x}'') \quad (2.23)$$

The visibility function that is normally part of the geometric term is not necessary since the obtained point is already the nearest surface point at the given direction.

Ray Tracing

To perform visibility tests and directional sampling, it is necessary to have a procedure to check for obstruction of the connection between two points and a procedure to search for a given starting point the nearest surface point at a given direction.

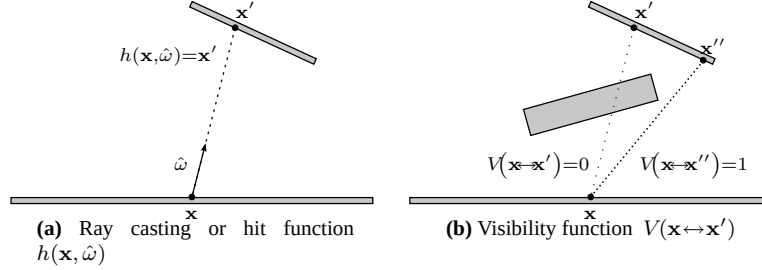


Figure 2.11: Basic Raytracing functions

A widespread algorithm that fulfills this requirements is *ray tracing*. We treat a ray tracing kernel as a black-box tool providing a *ray casting function* $h(\mathbf{x}, \hat{\omega})$ and a *visibility function* $V(\mathbf{x} \leftrightarrow \mathbf{x}')$. The *ray casting function* $h(\mathbf{x}, \hat{\omega})$ casts a ray from $\mathbf{x}$ into direction $\hat{\omega}$ and returns the nearest intersection point $\mathbf{x}'$ of the ray with the surface of the scene. The *visibility function* $V(\mathbf{x} \leftrightarrow \mathbf{x}')$, which returns zero if the segment $\overline{\mathbf{x} \mathbf{x}'}$ is blocked and one otherwise, can be emulated by the *ray casting function* but has its own optimized implementation in most cases.

A virtual scene has to be constructed from a set of geometric primitives. Very common primitives are polygons and parametric surfaces. Usually efficient ray tracing kernels check for a ray-scene intersection in two steps. First the ray is intersected with an acceleration structure as e.g. a *binary space partition* or a *bounding interval hierarchy*. Then all primitives that come into question of obstructing the ray are checked for intersection with the ray. Determining a ray-scene intersection by ray tracing with the use of a suitable acceleration structure is agreed to have a average complexity of order $O(\log N)$ for N primitives. More on this is found in the work of [Szirmay-Kalos and Márton, 1998].

2.3.2 Path Tracing

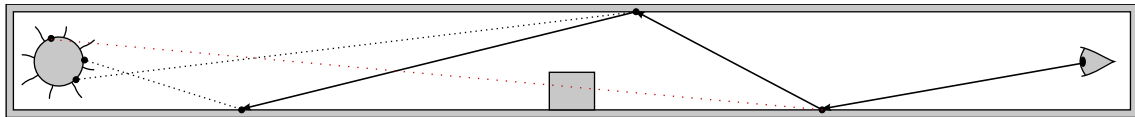


Figure 2.12: An exemplary path created by recursive Path Tracing with Next Event Estimation.

Path Tracing (PT) is a well known technique and is the naive approach to simulate a measurement when light transport is described as a recursively nested integral equation. Of course it is also compatible to the description of light transport as an integration problem over path space.

A path is generated in an unidirectional fashion by either starting at a sensor or starting at a light source and adding an edge to the path after every scattering event until the path is terminated through Russian roulette. By applying importance sampling according to the BSDF for path generation, the problem structure gets exploited in a greedy fashion but lacks any global considerations. In practice a primitive form of multiple importance sampling is often implicitly applied through partition: The direct lighting at any interaction point is captured by spatial sampling of the light sources and indirect lighting is captured by the continuation of the path by directional sampling. This technique is sometimes termed *Next Event Estimation*. To keep the two parts disjoint as necessary for the applied simple weighting strategy, light sources have to be treated like normal surfaces when hit through directional sampling. The unidirectional sample generation only

accounts for the characteristics of transport from the starting point. The characteristics of the transport to the target are neglected completely with the exception of the Next Event Estimation.

Path Tracing is a consistent and unbiased algorithm. It produces the correct image on average and converges for a growing number of samples. In practice it captures simple global illumination very well, but cannot efficiently handle difficult paths like that involved for caustics or lighting through a keyhole. Due to its simplicity and generality, Path Tracing is often used to calculate a reference solution.

2.3.3 Bidirectional Path Tracing

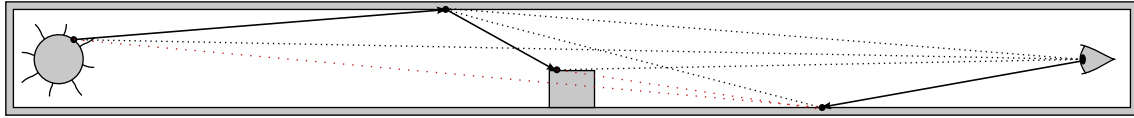


Figure 2.13: An exemplary path family created by Bidirectional Path Tracing.

Bidirectional Path Tracing (BDPT) has been developed by [Veach and Guibas, 1994] and [Lafortune and Willems, 1993] independently. [Veach] makes use of the Path Space as mathematical groundwork, because it facilitates an easy calculation of the sample density for a wide range of path generation strategies. For BDPT a path is generated by connecting two subpaths. One subpath is started at a light source, the other is started at a sensor and both are extended until termination by Russian roulette. Simple BDPT just connects the endpoints of the two subpaths. For full BDPT all possible connections are made to create a path family representing multiple samples. The different possible combinations of eye and light subpath length to obtain a path of any length are treated as distinct sampling techniques. Multiple importance Sampling is used to combine the different techniques.

A contributing path has to satisfy two conditions: of transporting light from the source and transporting light to the sensor. The two directions of path generation allow to account for the characteristics of light transport and its adjoint although only separately and not in combination. Only the connection edge depends on the spatial distribution of the interaction points from paths in both directions but does not account for the BSDF. The multiple importance sampling heuristics by Veach weight the members of a path families according to the combined characteristics of light transport.

Bidirectional Path Tracing is a consistent and unbiased algorithm. It produces the correct image on average and converges for an growing number of samples. In practice it captures simple and complex global illumination very well, but has high variance for extremely difficult configurations such as looking at a mirror that gives a view on caustics that are generated by a small light source shining on a specular surface.

2.3.4 Robust Instant Global Illumination

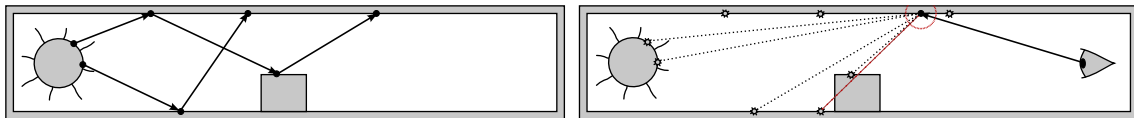


Figure 2.14: Exemplary subpaths from the shooting preprocess and the rendering process of Robust Instant GI.

The *Robust Instant Global Illumination* (IGI) algorithm by [Kollig and Keller, 2004] is a variant of Bidirectional Path Tracing, where light subpath interaction points created by a preprocess and stored for extensive reuse. The rendering process treats the stored interaction points like ordinary point lights with a directional emission characteristic. The same set of point lights is reused for every eye subpath.

The path generation is similar to BDPT, thus it shares the problems. The robust bounding heuristic used to robustly handle the weak singularity of the geometric term accounts for the local BSDF and determines the eye subpath length. The resultant *Robust Instant Global Illumination* (RIGI) algorithm handles specular

and glossy surfaces directly seen by the camera much better than the original IGI algorithm that is intended for purely diffuse scenes.

RIGI is a consistent and unbiased algorithm. It produces the correct image on average and converges for a growing number of samples. In practice it captures simple global illumination very well. For complex settings and highly specular surfaces, a very large number of stored light subpath interaction points is necessary. This nullifies the performance gain from extensive reuse through a extremely large number of necessary visibility tests. We propose a solution to this problem in chapter 3 by adaptively reducing the number of visibility tests. In chapter 4 a more general approach is attempted.

2.3.5 Metropolis Light Transport

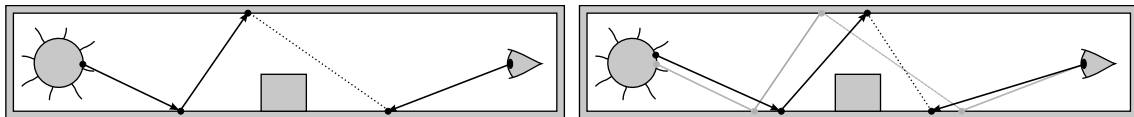


Figure 2.15: Two exemplary consecutive states from the markov chain of Metropolis Light Transport

The *Metropolis Light Transport* (MLT) algorithm by [Veach and Guibas, 1997] generates samples according to the density of light transport using an ergodic Markov chain whose stationary density is proportional to the image contribution function. This is achieved by choosing the transition probabilities of the Markov chain in a fashion that maintains detailed balance. The initial paths are generated using BDPT. New paths are generated from old paths by various mutation strategies. These paths are accepted with a probability that satisfies detailed balance. By creating samples according to the integrand, the probability density function and the integrated function cancel out. Thus all information on the integrand is encoded in the number of samples in the region of interest.

Through *two-pass MLT*, [Veach, 1997] even achieves an effect similar but not identical to a separation of the main part combined with importance sampling on the remaining differences. The energy densities estimated in a coarse first pass are used as reciprocal weights for the energy densities in the second pass. Thus a more uniform distribution of samples that concentrates in the regions of increased variance is achieved. Weighting with other characteristics of the coarse solution - like variance or entropy - might entail further improvements in the sample distribution.

Metropolis Light Transport combines the partial reuse of paths by mutations with sampling according to the energy density. Unfortunately those mutations make MLT a complicated algorithm and introduce additional requirements like the computability of conditional probability densities. Fortunately [Kelemen et al., 2002] found a way to reduce the MLT to the elegant quintessence of a sample generation mechanism which has no additional requirements to that of the underlying sampling algorithm, e.g. BDPT. In the current form this robust MLT variant has lost most of the original efficiency gains through path reuse.

Metropolis Light Transport is an algorithm which allows to sample path space according to the combined characteristics of light transport through clever probabilistic *rejection sampling*. It is a consistent and unbiased algorithm and produces the correct image on average and converges for a growing number of samples. In practice it captures complex global illumination better than the other presented algorithms but is often outperformed at easy settings where the density of light transport is piecewise dependent on mainly one factor. MLT has a reputation of being extremely complex, but we hope that this will change in the future, because in its core MLT is a simple and powerful algorithm.

2.3.6 Photon Mapping

Photon Mapping is an approximation technique published by [Jensen and Christensen, 1995] which shares some principles with BDPT. Both techniques join light subpaths and eye subpaths to generate complete transport paths. While BDPT completes paths by connecting the subpaths, Photon Mapping uses previously stored nearby light path vertices to approximate the outgoing radiance at the eye path vertex in question.

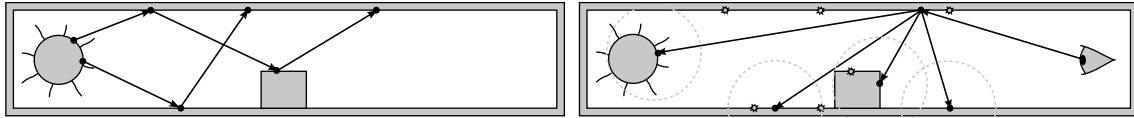


Figure 2.16: Exemplary subpaths from the shooting preprocess and the gathering process of Photon Mapping.

A more figurative view relates the algorithm to photon tracing. Photon packets are emitted from the light sources and traced through the scene. If a photon packet is scattered some amount of radiance is deposited and stored as a representative photon in a photon map. For the image generation this deposited radiance is gathered through a estimation of the photon density at the endpoints of eye subpaths. For a *density estimation* the photons in a certain neighborhood of the endpoint are acquired from the photon map to estimate the incoming radiance. The problem of finding the extremal paths associated with complex reflections is circumvented by approximating the contribution of those paths.

Photon Mapping is a consistent algorithm and converges to the correct image for a growing number of samples. The applied density estimation introduces some bias which disappears only for an infinite number of photons. It could be made unbiased at the cost of additional noise by using the technique presented in section 3.4. In practice Photon Mapping captures simple and complex global illumination very well, even difficult configurations that all other presented algorithms cannot handle robustly can be solved because of the approximative approach on the joining of subpaths.

Chapter 3

Adaptive Direct Illumination

In this chapter we concentrate on direct lighting by point light sources. Most lighting can be represented in this form. There are many techniques like those of [Cline et al., 2006] to create point lights from environment maps. The Robust Instant Global Illumination algorithm by [Kollig and Keller, 2004] converts path space samples into specialized point lights to represent direct and indirect lighting. The number of lights and the therefor necessary visibility tests are a limiting factor for the lighting complexity. In general, more complex scenes and lighting conditions require more point lights to adequately capture all lighting effects. Thus it is vital to compute the contribution and visibility of many lights efficiently. This can be achieved by reducing the number of visibility tests, but the evaluation of the BSDF, the geometric term, and other factors is also cost intensive. We construct an algorithm for efficient lighting evaluation that is simple, unbiased, scalable and requires minimal or no preprocessing. But first we survey previous work on similar issues.

[Ward, 1994] orders the light sources according to their potential contribution and then consecutively performs visibility tests, until an error bound dependent on pixel luminance is guaranteed. This deterministic error bound entails bias. The technique is inefficient for many lights with similar potential contribution in large portions of the image, which easily occurs for homogeneous lights, small scenes, or when using low frequency environmental lighting. Furthermore the per-pixel error bound based on luminance is difficult to satisfy in heavily shadowed areas. In exchange the algorithm is very simple and has very little overhead in processing time and memory requirements. The quality and error bound are easy to specify.

[Lafortune and Willems, 1995] perform one substitutional contribution calculation and visibility test for a whole group of light sources. The substitutional evaluation is based on the light source that is randomly chosen from a group according to its proportionate potential contribution. The algorithm is very simple, unbiased and has overhead comparable to that of the technique of [Ward]. It achieves an additional speedup by estimating the contribution of a whole group instead of evaluating the contribution of every single visible light. It does not use any error bound.

The basic strategy of *Light Cuts* by [Walter et al., 2005] is to replace the groups per light path, that were used by [Lafortune and Willems, 1995], with a tree structure to hierarchically represent light clustering. This adds some overhead for building, storing, and traversing the tree. Beginning at the root of the tree, the minimum and maximum contribution of a cluster are calculated. If a per-cluster error bound is satisfied only the representative light of the cluster is tested for visibility. Otherwise the evaluation recurses to the next hierarchy level. The per-cluster error bound preserves high contrast features, but can sum up per-pixel. Since the representative light or interaction point for a cluster is chosen when creating the tree structure, the resulting images differ considerably for different runs of the algorithm. *Multidimensional Light Cuts* by [Walter et al., 2006] extends the original algorithm with the clustering of the interaction points of eye subpaths.

[Wald et al., 2003] contrived a technique to reduce the light sources for *Interactive Global Illumination in Complex and Highly Occluded Environments*. The importance of the different light sources is determined using standard Path Tracing with sparse samples. All light sources with an importance below a certain threshold are discarded and the rest is sampled proportional to their relative importance. This technique works well in heavily occluded scenes and is unbiased when using a threshold of 0. Basing the sampling of

the light sources on their importance is beneficial for every scene.

[Keller and Wald, 2000] uses the connectivity information of stored light paths in the vicinity of the point of interest to select the set of the light sources that are most probably visible. The technique requires a range search to find nearby interaction points. This is sometimes problematic for sparse information. All probably visible light sources are tested for visibility. To avoid bias, the probably not visible light sources are evaluated using a very low sampling rate.

Applying Russian roulette like [Veatch, 1997] to the choice of performing or omitting a visibility test is a simple and unbiased procedure with very little overhead. A shortcoming is the increase of variance. [Szécsi et al., 2003] did some work on improving the efficiency of Russian roulette that has results similar to our work.

3.1 The Basic Framework

We begin the construction of our visibility test reduction technique with the outgoing radiance $\dot{L}_o(\mathbf{x}, \hat{\omega}_o)$ into direction $\hat{\omega}_o$ at point $\mathbf{x}$ due to the illumination from N point lights:

$$\dot{L}_o(\mathbf{x}, \hat{\omega}_o) = \sum_{n=1}^N C_n V_n \quad (3.1)$$

where:

N is the number of lights.

C_n is the potential contribution of the n^{th} light (i.e. the contribution if the light is visible).

V_n is the visibility of the n^{th} light.

If the point light sources are representatives of path space samples, as it is the case for Instant Global Illumination, the weighting by the reciprocal sample probability $\frac{1}{p_n}$ is included in the potential contribution C_n .

$$L_o(\mathbf{x}, \hat{\omega}_o) \approx \dot{L}_o(\mathbf{x}, \hat{\omega}_o) = \int_{\Omega} \mathbf{p}(\bar{\mathbf{x}}) \frac{c(\bar{\mathbf{x}})v(\bar{\mathbf{x}})}{p(\bar{\mathbf{x}})} d\mu(\bar{\mathbf{x}}) \approx \sum_{n=1}^N \frac{c(\bar{\mathbf{x}}_n)v(\bar{\mathbf{x}}_n)}{p(\bar{\mathbf{x}}_n)} = \sum_{n=1}^N \frac{1}{p_n} C'_n V_n = \sum_{n=1}^N C_n V_n \quad (3.2)$$

[Lafortune and Willems, 1995] use the proportionate potential contribution of lights to choose the one that is substitutionally tested for visibility.

$$\dot{L}_o(\mathbf{x}, \hat{\omega}_o) = \mathbf{E} \left(V_m \sum_{n=1}^N C_n \right) = \sum_{m=0}^N \mathbf{P}(\mathbf{m}) V_m \sum_{n=1}^N C_n \quad \text{with} \quad \mathbf{P}(\mathbf{m}) = \frac{C_m}{\sum_{k=1}^N C_k} \quad (3.3)$$

The *Russian roulette* based variant of this is to choose for every light if a visibility test is performed or the light is discarded completely according to an arbitrary probability $s_n \in (0, 1]$.

$$\dot{L}_o(\mathbf{x}, \hat{\omega}_o) = \mathbf{E} \left(\sum_{n=1}^N \begin{cases} \frac{1}{s_n} C_n V_n & \text{for } \xi_n \leq s_n \\ 0 & \text{otherwise} \end{cases} \right) = \sum_{n=1}^N (s_n \frac{1}{s_n} C_n V_n + (1 - s_n) \cdot 0) \quad (3.4)$$

Instead of discarding a light completely, we use an arbitrary predicted value for the visibility $\tilde{V}_n$. Obviously the quality of the result is best for correct visibility prediction. To avoid bias, a correction term has to be included.

$$\begin{aligned} \dot{L}_o(\mathbf{x}, \hat{\omega}_o) &= \mathbf{E} \left(\sum_{n=1}^N \begin{cases} \frac{1}{s_n} C_n (V_n - (1 - s_n) \tilde{V}_n) & \text{for } \xi_n \leq s_n \\ C_n \tilde{V}_n & \text{otherwise} \end{cases} \right) \\ &= \sum_{n=1}^N (s_n \frac{1}{s_n} C_n (V_n - (1 - s_n) \tilde{V}_n) + (1 - s_n) C_n \tilde{V}_n) \end{aligned} \quad (3.5)$$

This is simplified to:

$$\begin{aligned}\dot{L}_o(\mathbf{x}, \hat{\omega}_o) &= \mathbf{E} \left(\sum_{n=1}^N C_n \tilde{V}_n + \begin{cases} \frac{1}{s_n} C_n (V_n - \tilde{V}_n) & \text{for } \xi_n \leq s_n \\ 0 & \text{otherwise} \end{cases} \right) \\ &= \sum_{n=1}^N (C_n \tilde{V}_n + s_n \frac{1}{s_n} C_n (V_n - \tilde{V}_n) + (1 - s_n) \cdot 0)\end{aligned}\quad (3.6)$$

By setting s_n to the proportionate potential contribution $\frac{C_n}{\sum_{k=1}^N C_k}$ of a light, we apply importance sampling like [Lafortune and Willems, 1995](see (3.3)).

$$\begin{aligned}\dot{L}_o(\mathbf{x}, \hat{\omega}_o) &= \mathbf{E} \left(\sum_{n=1}^N C_n \tilde{V}_n + \begin{cases} (\sum_{k=0}^N C_k) (V_n - \tilde{V}_n) & \text{for } \xi_n \leq s_n \\ 0 & \text{otherwise} \end{cases} \right) \\ &= \sum_{n=1}^N (C_n \tilde{V}_n + s_n (\sum_{k=0}^N C_k) (V_n - \tilde{V}_n) + (1 - s_n) \cdot 0)\end{aligned}\quad (3.7)$$

We introduce the quality factor Q_n to scale the basic probability of making a visibility test and modify it according to our confidence in the visibility prediction and other additional information. Because Q_n has to conform to the requirement $s_n Q_n \in (0, 1]$ the value for Q_n is bounded by $(0, Q_n \leq \frac{\sum_{k=1}^N C_k}{C_n}]$. To ensure this condition we set

$$Q_n = q_n \frac{\sum_{k=1}^N C_k}{C_n} \quad \text{with} \quad q_n \in (0, 1]. \quad (3.8)$$

The final estimator corresponding to our visibility test reduction method is

$$\begin{aligned}\ddot{L}_o(\mathbf{x}, \hat{\omega}_o) &= \mathbf{E} \left(\sum_{n=1}^N C_n \tilde{V}_n + \begin{cases} \frac{1}{Q_n} (\sum_{k=1}^N C_k) (V_n - \tilde{V}_n) & \text{for } \xi_n \leq s_n Q_n \\ 0 & \text{otherwise} \end{cases} \right) \\ &= \sum_{n=1}^N (C_n \tilde{V}_n + s_n Q_n \frac{1}{Q_n} (\sum_{k=1}^N C_k) (V_n - \tilde{V}_n) + (1 - s_n Q_n) \cdot 0).\end{aligned}\quad (3.9)$$

Now we have got a simple method to reduce the number of visibility tests. Since final estimator of our method is a equivalent transformation of the original unbiased estimator, it is unbiased itself. In order to apply importance sampling in the proposed way, the potential contribution of each light source has to be calculated or estimated. Furthermore a method to predict the visibility is necessary.

3.1.1 Efficiency

The variance $V(\dot{L}_{o,N})$ from a underlying approximation of the illumination by N point lights is

$$V(\dot{L}_{o,N}) = \frac{1}{N} V(f(\bar{x})) = \sum_{n=1}^N V\left(\frac{1}{N} f(\bar{x}_n)\right) = \sum_{n=1}^N V(C_n V_n) = \frac{1}{N^2} \sum_{n=1}^N V(\dot{C}_n V_n), \quad (3.10)$$

where the contribution C_n can be factorized to $C_n = \frac{1}{N} \dot{C}_n$. If more lights are used to approximate the illumination, the expected result is a lower variance.

The variance when attempting a reduction of the visibility tests with our method is given by:

$$V(\ddot{L}_o(\mathbf{x}, \hat{\omega}_o)) = \sum_{n=1}^N \mathbf{E} \left(\left(\frac{\sum_{k=1}^N C_k}{Q_n C_n} - 1 \right) C_n^2 (V_n - \tilde{V}_n)^2 \right) + \frac{1}{N^2} \sum_{n=1}^N V(C_n V_n). \quad (3.11)$$

$$V(\ddot{L}_o(\mathbf{x}, \hat{\omega}_o)) = \sum_{n=1}^N \mathbf{E} \left(\left(\frac{\frac{1}{N} \sum_{k=1}^N \dot{C}_k}{Q_n \frac{1}{N} \dot{C}_n} - 1 \right) \frac{1}{N^2} \dot{C}_n^2 (V_n - \tilde{V}_n)^2 \right) + \frac{1}{N} V(\dot{C}_n V_n). \quad (3.12)$$

The additionally introduced variance $V^+(\ddot{L}_{o,N})$ is dependent on the deviation of the predicted value from the correct value:

$$V^+(\ddot{L}_{o,N}) = V(\ddot{L}_o(\mathbf{x}, \hat{\omega}_o)) - V(\dot{L}_{o,N}) = \frac{1}{N^2} \sum_{n=1}^N E \left(\left(\frac{\frac{1}{N} \sum_{k=1}^N \dot{C}_k}{\frac{1}{N} Q_n \dot{C}_n} - 1 \right) \dot{C}_n^2 (V_n - \tilde{V}_n)^2 \right). \quad (3.13)$$

The additional variance due to a single point light is composed of two factors: One factor is the reciprocal of the fraction of executed visibility tests times the relation of the contribution of the current light to the average contribution, all together minus 1. The other factor is the squared deviation of the predicted value from the correct value. The mean of Q_n is the expected number of visibility tests. The mean of $\frac{Q_n}{N}$ consequently equals the proportional fraction of executed visibility tests.

If the cost of a visibility test is T_V , the cost of a single visibility prediction is $T_{\tilde{V}}$ and the cost of a contribution evaluation is T_C , we can calculate the efficiency as follows:

$$\epsilon^+ = \frac{1}{(V(\dot{L}_{o,N}) + V^+(\ddot{L}_{o,N}))(NT_{\tilde{V}} + NT_C + (\sum_{n=1}^N \frac{q_n}{N})T_V)} \quad (3.14)$$

The efficiency of a full evaluation of all point lights is given by:

$$\epsilon = \frac{1}{(V(\dot{L}_{o,N}))(NT_C + NT_V)} \quad (3.15)$$

Since the variance is increased and the cost is decreased by our technique the aimed for decrease of variance has to be the result of an increased number of samples.

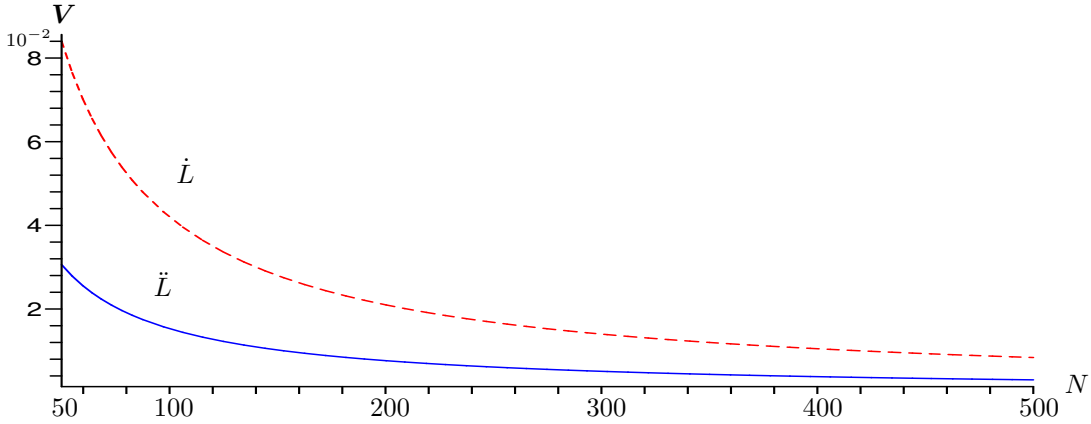


Figure 3.1: The solid curve shows the variance of our estimator for a number of light sources $N \in [50, 500]$. For comparison the dashed curve shows the variance of a full evaluation for the number of light sources that results in equal costs. We assume a variance of 2 for $\dot{C}$, a quality of the guesses resulting in $(V_n - \tilde{V}_n)^2 = 0.01$, and set the costs to $T_V = 20$, $T_{\tilde{V}} = 1$ and $T_C = 1$.

3.2 Prediction of Visibility and Contribution

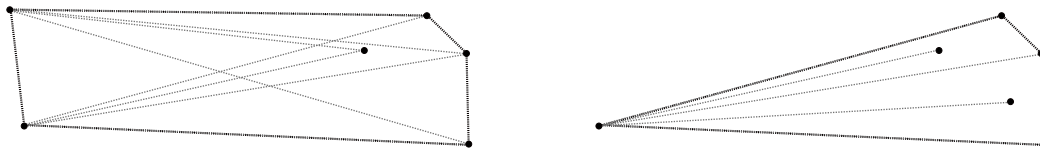
The variance of our technique depends on the similarity of the predicted value and the real value for visibility or contribution, so we need a good prediction of the respective value. Apart from dedicated prediction techniques, any illumination approximation technique can be used to predict the visibility and contribution of a light source.

The confidence in the prediction is another important factor. If we are confident that the predicted value is correct, we can use a low correction probability and hence few visibility tests. If the variance of the true visibility or contribution is high, a prediction is unreliable and consequently many visibility tests are required. For a small maximum possible error, a lower confidence level than for a big maximum possible error is sufficient.

3.2.1 Coherency in Illumination

Coherence in illumination backs the use of a predicted value that is based on nearby samples instead of a full evaluation. Realistic scenes feature regions of coherent illumination, but also illumination discontinuities and erratic illumination effects. To capture the high frequency effects of illumination, we naturally need more samples than for the smooth low frequency effects.

The coherency in illumination is caused by the coherency of light transport paths. The space that is traversed by transport paths between two groups of points can be bounded by the polygonal object spanned by all points. The respective object resembles a pyramid if one group consists of a single point. If the points span a small region of space, the visibility test for a single light source have to check nearly the same space for obstruction and thus are likely to have similar results.



(a) A bound of the space traversed for connections between a group of points at the left side and a group of points at the right side

(b) A bound of the space traversed for connections between a single point at the left side and a group of points at the right side

Figure 3.2: Bounds on the space traversed for connections between groups of points.

Usually there are much more samples used for image generation than there are light sources in the scene, so the primary source of points with coherent location are the first few interaction points of eye paths. For very many lights coherency starts to show for light path interaction points, too. Due to scattering, the coherency of the points rapidly decreases with increasing path length. High frequency features like object boundaries or focused scattering also break the coherency. Coherency of the contribution of light sources requires not only coherency of the location, but also coherency of orientation and coherency in surface properties.

3.2.2 Visibility Prediction

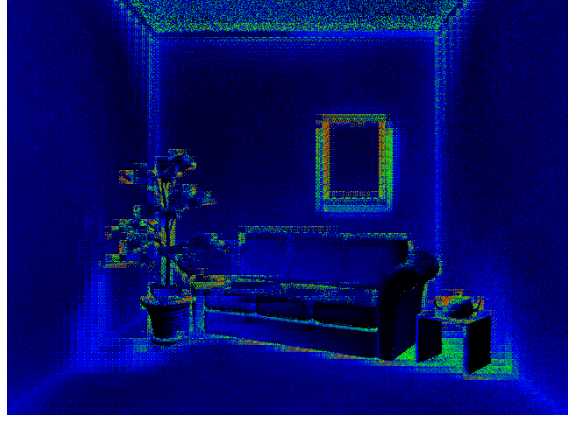
Using a constant to predict the visibility of a light is extremely cheap, but results in satisfying prediction quality only for a sensible choice of the constant. The lack of any dependency on location is problematic since there may be large regions, where the average visibility is close to 0, and others, where the average visibility is close to 1. A guess of about 0.5 does not fit very well in either region. Typically the direct illumination is not coherent for the whole scene and thus a constant bound is not reasonable.

Another simple approach on visibility prediction is to use the averaged result of all previously performed visibility tests for a light like [Ward, 1994]. The required statistics build up very fast and automate the choice of the value for a constant visibility prediction. The value has no direct dependency on the location, but the evaluation order induces some dependency on the location and thus improves the guess. This effect can be amplified by using a running average and evaluating the pixels in the order given by a space filling curve on the image plane. Alternatively the results from visibility tests from *nearby* samples can be averaged. The definition of *nearby* can depend on various properties like the adjacency of pixels, the depth value, the surface orientation, the location of the query point, the occupied grid cell, and much more. A smaller area of regard for nearby samples usually produces a better fit for high frequency effects, but also reduces the number of available visibility samples, which in turn increases the variance of the guess. High variance causes low confidence, but nevertheless a smaller area of regard is reasonable for a higher sampling rate.

We have tried out various criteria for spatial grouping to determine near samples. Examples of groups are all samples from a pixel tile, all samples in the same spatial grid cell, all samples acquired by a range search, all samples from a local part nearby scan lines, and many others. The scan line based approach worked very well and required only a small constant amount of memory.

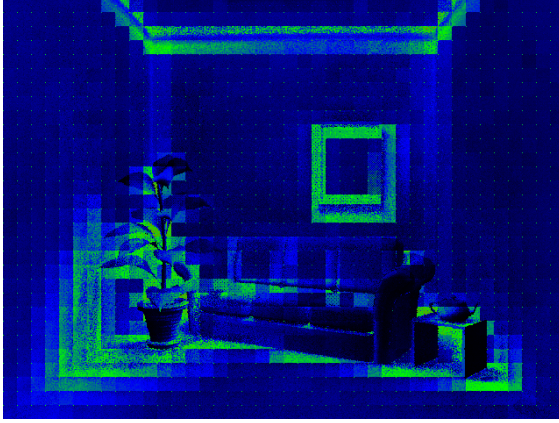


(a) This image rendered with a single random eye subpath sample per pixel exhibits smooth shadows and an low noise level.

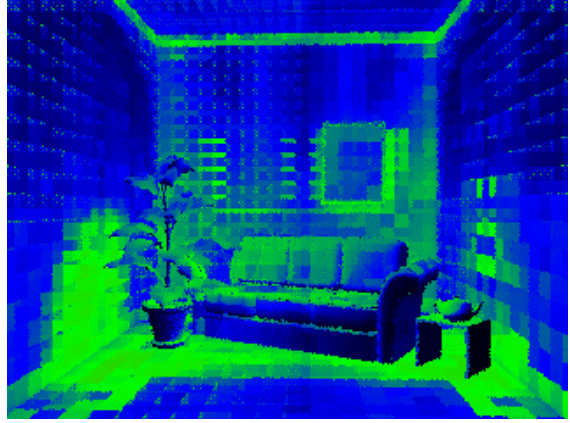


(b) Color encoded number of shadow tests. Black denotes no tests, black to blue up to $\frac{1}{3}$, blue to green up to $\frac{2}{3}$ of all possible tests and green to red up to all tests

Figure 3.3: Images generated with an proof-of-concept implementation of the proposed shadow test reduction technique. Of the 2482 point light sources 17% are have been tested for visibility.



(a) 16×16 pixel tiles



(b) Spatial grid cells

Figure 3.4: The color encoding of the number of shadow tests reveals the underlying spatial grouping.

3.2.3 Contribution Prediction

Evaluating some parts of the contribution C_n may be costly. We can use the technique for the reduction of visibility tests to avoid the full evaluation of the contributions from unimportant light sources. We factorize C_n into $\hat{D}_n E_n F_n$. To determine s_n the factor D_n is always evaluated and E_n is estimated by $\tilde{E}_n$. F_n is only used for visible lights. This leads to:

$$\begin{aligned} \ddot{L}_o(\mathbf{x}, \hat{\omega}_o) &= E \left(\sum_{n=0}^N D_n \tilde{E}_n \tilde{V}_n \begin{cases} \frac{1}{Q_n} (\sum_{k=1}^N D_k \tilde{E}_k) (E_n F_n V_n - \tilde{E}_n \tilde{V}_n) & \text{for } \xi_n \leq s_n Q_n \\ 0 & \text{otherwise} \end{cases} \right) \\ &= \sum_{n=1}^N (D_n \tilde{E}_n F_n \tilde{V}_n + s_n Q_n \frac{1}{Q_n} (\sum_{k=1}^N D_k \tilde{E}_k) (E_n F_n V_n - \tilde{E}_n \tilde{V}_n) + (1 - s_n Q_n) \cdot 0) \quad (3.16) \end{aligned}$$

$$\text{where } s_n = \frac{D_n \tilde{E}_n}{\sum_{k=0}^N D_k \tilde{E}_k} \text{ and } s_n Q_n \in (0, 1]$$

3.3 Unbiased Light Cuts

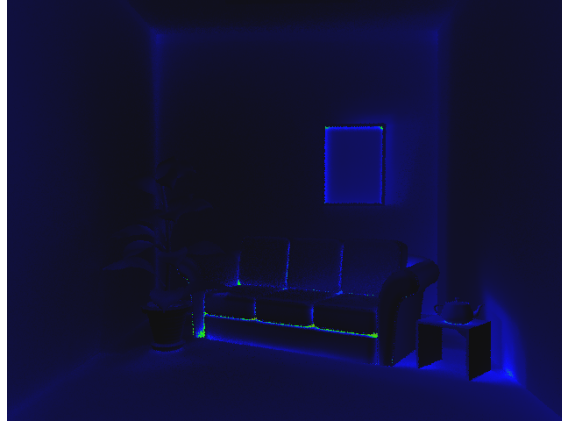
For very many light sources even the contribution evaluation and the random decision per light source sum up to significant costs. To handle many light sources efficiently we have to break the linear dependency on the number of light source. Hierarchical structures allow for evaluations with costs sub-linear to the number of elements.

Light Cuts by [Walter et al., 2005] describes an algorithm to reduce the shadow tests in scenes with many lights, where the light sources are hierarchically grouped to clusters. The clusters are evaluated top-down. If a fixed threshold for the maximum error per cluster is undershot, further refinement of the current branch of the hierarchy is stopped and the proportionate contribution of a representative light source of the current refinement level is reused for all the remaining light sources of the branch. The use of a fixed error threshold preserves high contrast features, but for the complete hierarchy the maximum possible error sums up.

The decision based on a threshold for the conservatively estimated cluster error resembles *importance sampling* according to the maximum possible error. Unfortunately the deterministic nature of the decision on refinement entails bias. To remove the bias, we randomize the decision on refinement by *Russian roulette*. The estimated contribution of a cluster is used to reduce the additional variance introduced by Russian roulette. The inherent unbiasedness of this procedure allows to predict the illumination from a whole cluster of light sources by any prediction technique. In addition the randomization dissolves artifacts due to differences in the used hierarchy levels to noise.



(a) The resulting color image exhibits very smooth shadows and a low noise level. The image was rendered with a single random eye subpath sample per pixel.



(b) Color encoded number of shadow tests. Black denotes no tests, black to blue up to $\frac{1}{3}$, blue to green up to $\frac{2}{3}$ of all possible tests and green to red up to all tests

Figure 3.5: Images generated with an proof-of-concept implementation of the proposed shadow test reduction technique using hierarchical clustering and contribution prediction. The decision on descent is based on the confidence into the prediction. Of the 24980 point light sources on average 1300 are tested for visibility in every pixel. This equals a reduction to 12% of all necessary visibility tests.

3.4 Predictions with Correction at Random

Although developed for the purpose of reducing visibility tests, our basic framework can be applied to more general problems. An equivalence transformation of the expected value of the primary Monte Carlo estimator reveals the similarity of our technique to a combination of *Russian roulette* and the *separation of a constant main part* $\tilde{m} \equiv \frac{\tilde{M}}{\mu(\Omega)}$:

$$F = E \left(\frac{f(x)}{p(x)} \right) = \int_{\Omega} \mathbf{p}(\mathbf{x}) \frac{f(x)}{p(x)} d\mu(x) \quad (3.17a)$$

$$= E \left(\begin{cases} \frac{1}{p} \frac{f(x)}{p(x)} & \text{for } \xi \leq \mathbf{p} \\ 0 & \text{else} \end{cases} \right) = \mathbf{p} \frac{1}{p} \int_{\Omega} \mathbf{p}(\mathbf{x}) \frac{f(x)}{p(x)} d\mu(x) + (1 - \mathbf{p}) \cdot 0 \quad (3.17b)$$

$$= E \left(\begin{cases} \frac{1}{p} \left(\frac{f(x)}{p(x)} - \tilde{M} \right) & \text{for } \xi \leq \mathbf{p} \\ 0 & \text{otherwise} \end{cases} \right) + \tilde{M} = \mathbf{p} \frac{1}{p} \left(\int_{\Omega} \mathbf{p}(\mathbf{x}) \frac{f(x)}{p(x)} d\mu(x) - \tilde{M} \right) + \tilde{M} \quad (3.17c)$$

$$= E \left(\begin{cases} \frac{1}{p} \frac{f(x) - \tilde{m}}{p(x)} & \text{for } \xi \leq \mathbf{p} \\ 0 & \text{otherwise} \end{cases} \right) + \tilde{M} = \mathbf{p} \frac{1}{p} \left(\int_{\Omega} \mathbf{p}(\mathbf{x}) \frac{f(x) - \tilde{m}}{p(x)} d\mu(x) \right) + \tilde{M} \quad (3.17d)$$

Since the value of $\tilde{M}$ has no effect on the expected value, it can be chosen arbitrarily without introducing bias. Thus it is perfectly valid to use the previously calculated result of another estimator for the main part, although the final estimator is then correlated to the one used for the main part. The probability p has to be non-zero to .

The proposed algorithm is a versatile tool to replace costly evaluations by a cheap prediction while ensuring unbiasedness. The usual bias is replaced by noise caused by the Russian roulette based part of the algorithm. The magnitude of the noise depends on the quality of the prediction and the correction probability. Predicted values that differ greatly from the real value and a low correction probability increase the noise level.

3.4.1 Unbiased Adaptive Sampling

One application of *prediction with correction at random* is adaptive sampling. Adaptively varying the number of samples for an estimator easily introduces bias if not done carefully. Consider a checkerboard that is placed so that one half of each pixel is white and the other half is black. A first estimator is used to estimate all pixel values. Some are darker and some are brighter than the expected average. If we use this information to decide on the use of an additional estimator, we easily introduce bias. For example, if we use the second estimator in the case of a pixel that is brighter than some threshold, the affected pixel values most likely get nearer to the correct solution, which means darker in that case. Therefore the whole image gets darker on average and thus is biased. Refer to the work of [Kirk and Arvo, 1991] for some further reasoning on that topic.

We can use the combination of Russian roulette with the separation of a constant main part for an estimator with a variable number of samples. The exact number of samples is subject to randomization but the mean number of samples can be specified through the threshold of Russian roulette.

We begin our construction with a linear convex combination of Monte Carlo estimators. As long as every used estimator is unbiased, the final estimator is unbiased, too, since the expected value is linear. We use a primary estimator as defined in equation (3.17c) with $\frac{\tilde{F}_n}{|\tilde{\Omega}|} = \tilde{f}_n$ as main part and an arbitrary function $\tilde{f}_0$ to

set up our estimator. After that we extend it to any number of samples N :

$$\begin{aligned}
 F &= E(\tilde{F}_1) = \mathbf{p}_1 \frac{1}{p_1} \left(\int_{\Omega} f(x) d\mu(x) - E(\tilde{F}_0) \right) + E(\tilde{F}_0) \\
 F &= E(\tilde{F}_2) = \frac{1}{2} \left[E(\tilde{F}_1) + \mathbf{p}_2 \frac{1}{p_2} \left(\int_{\Omega} f(x) d\mu(x) - E(\tilde{F}_1) \right) + E(\tilde{F}_1) \right] \\
 F &= E(\tilde{F}_N) = \frac{1}{N} \left[E(\tilde{F}_1) + E(\tilde{F}_2) + \dots + \mathbf{p}_N \frac{1}{p_N} \left(\int_{\Omega} f(x) d\mu(x) - E(\tilde{F}_{N-1}) \right) + E(\tilde{F}_{N-1}) \right] \\
 &= \frac{1}{N} \left[\sum_{n=1}^N \left(\mathbf{p}_n \frac{1}{p_n} \left(\int_{\Omega} f(x) d\mu(x) - E(\tilde{F}_{n-1}) \right) + E(\tilde{F}_{n-1}) \right) \right] \tag{3.18}
 \end{aligned}$$

We may use any unbiased estimator to evaluate the target integral and still retain unbiasedness. We denote the expected value of the applied estimator by $\int_{\Omega} f(x) d\mu(x) \approx E(\tilde{e}(f))$.

$$F = \int_{\Omega} f(x) d\mu(x) = E(\tilde{F}_N) = \frac{1}{N} \left[\sum_{n=1}^N \left(\mathbf{p}_n \frac{1}{p_n} \left(E(\tilde{e}(f)) - E(\tilde{F}_{n-1}) \right) + E(\tilde{F}_{n-1}) \right) \right] \tag{3.19}$$

Further optimizations result e.g. from nesting the Russian roulette based random decisions.

$$E(\tilde{F}_N) = \frac{1}{N} \left[\mathbf{p}_1 \frac{1}{p_1} \left(E(\tilde{e}(f)) - E(\tilde{F}_0) + \left[\mathbf{p}_2 \frac{1}{p_2} \left(E(\tilde{e}(f)) - E(\tilde{F}_1) + [\dots] \right) + E(\tilde{F}_1) \right] \right) + E(\tilde{F}_0) \right] \tag{3.20}$$

The corresponding estimator is defined recursively:

$$F \approx \tilde{F}_n = \frac{1}{n} \left[(n-1) \tilde{F}_{n-1} + \left\{ \left(\prod_{k=0}^{n-1} \frac{1}{p_k} \right) \left[\begin{cases} \frac{1}{p_n} (\tilde{e}(f) - \tilde{F}_{n-1}) & \text{for } \xi_n \leq p_n \\ 0 & \text{else} \end{cases} \right] + \tilde{F}_{n-1} \right\} \text{ for } \xi_k \leq p_k \Big|_{k=0}^{n-1} \right] \text{ otherwise } \tag{3.21}$$

Finally we conceived an unbiased estimator with an adaptively chosen number of samples. The randomized number of samples is dependent on the choice of the probabilities p_n . If we base the probability p_n for using the estimator $\tilde{F}_n$ on observations from earlier samples like e.g. the difference of $\tilde{F}_{n-1}$ and $\tilde{F}_{n-2}$, it correlates the estimators, but still does not introduce bias, because a compensation for any p_n is an intrinsic part of the complete estimator. In principle, the whole estimator is a predetermined convex combination of estimators where some estimators are omitted at random.

The fundamental difference between our adaptive sampling strategy and traditional adaptive sampling is the randomization of the refinement decision. This distinction is comparable to that of *quasi-Monte Carlo integration* from *randomized quasi-Monte Carlo integration*. It bridges the gap between a consistent estimator and an unbiased consistent estimator. It has to be noted that the induced correlation can increase the variance of the estimator under some circumstances. A poor choice of the correction probability can further increase the variance and amplify artifacts created by correlation. For practical use, the storage required for the main part has to be balanced against the quality and distinctness of the main part.

Efficiency

The efficiency of an estimator is defined as $\epsilon = \frac{1}{\sqrt{V}}$ where V is the variance of the used estimator and T the cost for the evaluation of the estimator.

If $\tilde{F}_{n-1}$ and $\tilde{e}(f)$ are independent and the expected values of $\tilde{F}_{n-1}$ and $\tilde{e}$ match, the variance of the extended estimator $\tilde{F}_N$ is

$$V[\tilde{F}_n] = V[\tilde{F}_{n-1}] + \frac{1}{n^2} \left(\left(\frac{1}{p} - 2n \right) V[\tilde{F}_{n-1}] + \frac{1}{p} V[\tilde{e}(f)] \right). \tag{3.22}$$

The extension of the estimator $\tilde{F}_{n-n}$ to the estimator $\tilde{F}_n$ reduces the variance of the estimator if

$$p \geq \frac{1 + \frac{V[\tilde{e}(f)]}{V[\tilde{F}_{n-1}]}}{2n}. \quad (3.23)$$

The efficiency of the estimator $\tilde{F}_{n-1}$ with a variance of $V(\tilde{F}_{n-1})$ and a cost of $T(\tilde{F}_{n-1})$ is:

$$\epsilon_{\tilde{F}_{n-1}} = \frac{1}{V(\tilde{F}_{n-1})T(\tilde{F}_{n-1})}. \quad (3.24)$$

The efficiency of the extended estimator $\tilde{F}_n$ is

$$\epsilon_{\tilde{F}_n} = \frac{1}{\left(V[\tilde{F}_{n-1}] + \frac{1}{n^2} \left(\left(\frac{1}{p} - 2n \right) V[\tilde{F}_{n-1}] + \frac{1}{p} V[\tilde{e}(f)] \right) \right) \left(T(\tilde{F}_{n-1}) + T(\tilde{e}) \right)}. \quad (3.25)$$

The efficiency of the extended estimator $\tilde{F}_n$ is higher than the efficiency of the unextended estimator $\tilde{F}_{n-1}$ if

$$\epsilon_{\tilde{F}_{n-1}} \leq \epsilon_{\tilde{F}_n} \quad (3.26a)$$

$$p \geq \frac{\left(V(\tilde{F}_{n-1}) + V(\tilde{e}) \right) \left(T(\tilde{F}_{n-1}) + T(\tilde{e}) \right)}{2nV(\tilde{F}_{n-1}) \left(T(\tilde{F}_{n-1}) + T(\tilde{e}) \right) - n^2V(\tilde{F}_{n-1})T(\tilde{e})}. \quad (3.26b)$$

Chapter 4

Adaptive Global Illumination

The Robust Instant Global Illumination algorithm is a very efficient at capturing low frequency global illumination. In this chapter we briefly recapitulate the algorithm. We extend the range of application of RIGI by some extensions that effect a better adaptation of the sample distribution to the distribution of important light transport paths.

The original core of the *Robust Instant Global Illumination* algorithm has been developed by [Keller, 1997] as a radiosity method without a finite element kernel or solution discretization and was thus named *Instant Radiosity*. The Bidirectional Path Tracing interpretation of the algorithm was revealed with the introduction of the bound for the geometric term proposed in *Illumination in the Presence of Weak Singularities* by [Kollig and Keller, 2004]. We prefer the BDPT interpretation of the algorithm because of its generality. This interpretation allows for a clear view on the advantages and weak spots of the algorithm.

In principle RIGI is a very clever variant of BDPT that employs extensive reuse. The generation of the light subpaths and the eye subpaths is handled by different parts of the algorithm. The light subpaths are generated in the same fashion as for BDPT, but in a preprocess. Each light subpath is started by sampling a point on any light source specified in the scene description. Then in turn the next interaction point with the scene is sampled and Russian roulette is applied to decide upon the termination of the path. Instead of combining the created light subpath with an eye subpath directly, all light subpaths are stored for later reuse. The eye subpaths are generated almost the same way but start at the sensor. Every interaction point of the eye subpath is connected with each of the stored light subpath interaction points to construct multiple complete light transport paths.

As with BDPT, the different possible combinations of eye and light subpath length to obtain a path of any length are treated as distinct sampling techniques. Multiple importance sampling is used to combine them. The Instant Radiosity algorithm only uses eye paths with one edge to keep the path space regions that are sampled by different techniques disjoint. For Robust Instant Global Illumination the *robust bounding heuristic* is used calculate the weights for the individual subpath combinations.

A principal advantage of *RIGI* is the high performance thanks to extensive reuse. A comparatively low number of light subpaths is generated in the preprocess for negligible costs. Only the costs of the generation of the eye subpaths and the costs for the visibility tests of the deterministic connections to the light subpaths are significant. The repeated use of the same light paths makes the illumination look smooth and convincing, even if the sampling rate indicates that the result is probably still far from convergence. Much too low sampling rates result in artifacts like multiple sharp shadow edges where a smooth shadow would be correct.

Robust Instant Global Illumination algorithm also has some weak spots:

The Insufficient Sampling Rate for High Frequency Illumination Effects is a problem directly entailed to the extensive reuse of the light subpaths. It cannot be solved by increasing the number of the light subpaths, because this approach eventually overcompensates the performance gain from reuse by the increased number of required visibility tests. An existing technique to reduce the problem is *Interleaved Sampling* by [Source]. Its is successful in hiding the deficiencies of a single set of light subpaths by interleaving images generated with different sets, but the individual sets exhibit the same

problem at their respective image parts. The affected illumination effects — like for example caustic patterns — are quite common in realistic scenes and should be handled robustly. In the original paper it is proposed to omit this kind of paths and apply Photon Mapping to capture the high frequency illumination effects.

The Imperfect Sample Distribution is a general problem of path tracing algorithms. A bidirectional path tracing strategy based on local sampling techniques cannot simultaneously adapt to the characteristics of light transport and its adjoint without special provisions like that of [Jensen, 1995].

The principal effect from imperfect importance sampling is a sample distribution that is not fitting the density of light transport. This is most noticeable if samples are missing from regions with high contribution and high variance. The results typically are severe noise and artifacts for high frequency indirect illumination effects.

4.1 Bounds for the Geometric Term

The bound used by the robust bounding heuristic has a high influence on the course and length of the eye subpaths. Therefore its structure is a key factor for a advantageous sample distribution. The initial reasons for bounding the geometric term are of a completely different nature: The contribution due to a deterministic connection of an eye subpath interaction point with an light subpath interaction point has to include the geometric term. The geometric term has the distance between the two connected points as its denominator. As this distance approaches zero, the Geometric Term approaches infinity. This singularity causes infinite variance. To avoid the infinite variance introduced by the weak singularity of the geometric term, it is common practice to bound the geometric term.

In images generated with algorithms similar to Instant Radiosity, the effects of the weak singularity of the geometric term are visible as bright spots on concave surfaces near the reused light subpath interaction points. This kind of artifact has been named *blossoming* by [Kollig and Keller, 2004].

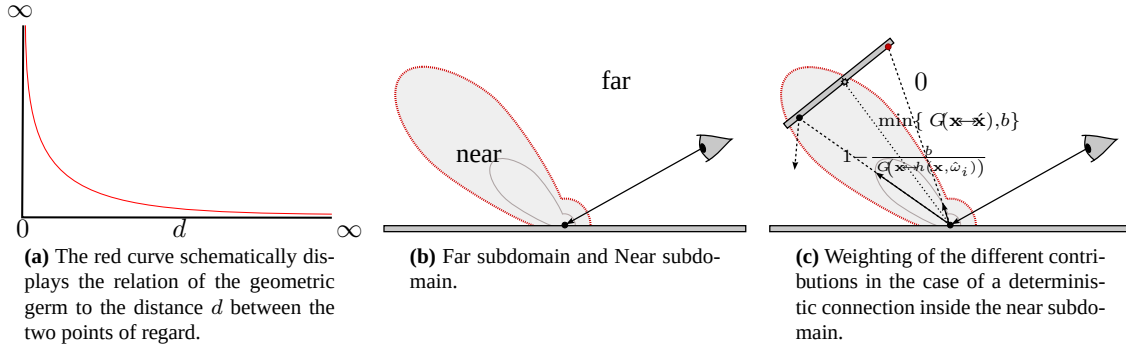


Figure 4.1: Why and how to bound the Geometric Term.

[Kollig and Keller, 2004] have been the first to develop an elegant compensation term for the bias introduced by bounding the geometric term. In effect the integration domain is decomposed into a near subdomain and a far subdomain.

$$L(\mathbf{x} \rightarrow \dot{\mathbf{x}}) = L_e(\mathbf{x} \rightarrow \dot{\mathbf{x}}) + \int_{\mathcal{M}} f_s(\dot{\mathbf{x}} \rightarrow \mathbf{x} \rightarrow \dot{\mathbf{x}}) L(\dot{\mathbf{x}} \rightarrow \mathbf{x}) \min\{G(\mathbf{x} \leftrightarrow \dot{\mathbf{x}}), b\} dA(\mathbf{x}) \quad (4.1a)$$

$$+ \int_{S^2} f_s(\hat{\omega}_o, \mathbf{x}, \hat{\omega}_i) L_o(h(\mathbf{x}, \hat{\omega}_i), -\hat{\omega}_i) \frac{\max\{G(\mathbf{x} \leftrightarrow h(\mathbf{x}, \hat{\omega}_i)) - b, 0\}}{G(\mathbf{x} \leftrightarrow h(\mathbf{x}, \hat{\omega}_i))} d\sigma_{\mathbf{x}}^{\perp}(\hat{\omega}_i) \quad (4.1b)$$

If the light subpath interaction point is outside the near subdomain that is anchored at the eye subpath interaction point, the deterministic connection is handled as usual. If it is inside the near subdomain, the geometric term from the deterministic connection is bounded by b . The bounding is compensated by

extending the eye subpath. The contribution of the extended path is weighted by $\left(1 - \frac{b}{G(\mathbf{x} \leftrightarrow h(\mathbf{x}, \hat{\omega}_i))}\right)$ if $G(\mathbf{x} \leftrightarrow h(\mathbf{x}, \hat{\omega}_i)) > b$ and otherwise is discarded.

This bounding mechanism can be reformulated as an heuristic for multiple importance sampling to seamlessly fit into the BDPT framework. The different combinations (l_e, l_l) of eye subpath length l_e and light subpath length $l_l = l - l_e$ to generate a path of the length l are treated as different sampling techniques. The weights for the individual techniques are defined as follows:

$$\begin{aligned}
 w_{(0,l-0)}(\bar{x}) &= \min\left\{1, \frac{b}{G(\mathbf{x}_0 \leftrightarrow \mathbf{x}_1)}\right\} \\
 w_{(1,l-1)}(\bar{x}) &= \min\left\{1, \frac{b}{G(\mathbf{x}_1 \leftrightarrow \mathbf{x}_2)}\right\} \cdot (1 - w_{(0,l-0)}) \\
 &\vdots \\
 w_{(l-1,1)}(\bar{x}) &= \min\left\{1, \frac{b}{G(\mathbf{x}_{l-1} \leftrightarrow \mathbf{x}_l)}\right\} \cdot (1 - w_{(l-1,1)}) \cdots (1 - w_{(0,l-0)}) \\
 w_{(l-0,0)}(\bar{x}) &= (1 - w_{(l-1,1)}) \cdots (1 - w_{(0,l-0)})
 \end{aligned} \tag{4.2}$$

Any non-negative bound b results in an unbiased estimator, but the impact on efficiency and image quality varies. For a low bound the near subdomain is large and that way an increased fraction of contribution is due to the extension of eye subpaths. Blossoming is reduced as the average eye subpath length increase, but the overall cost also increases due to the additional ray casts. A high bound results in a small near subdomain and thus reduces the costs for eye subpath, but blossoming may occur as an unwanted side effect. Since it is the geometric term which is bounded, the bound is closely linked to the scale of the scene and thus the bound has to be adjusted appropriately.

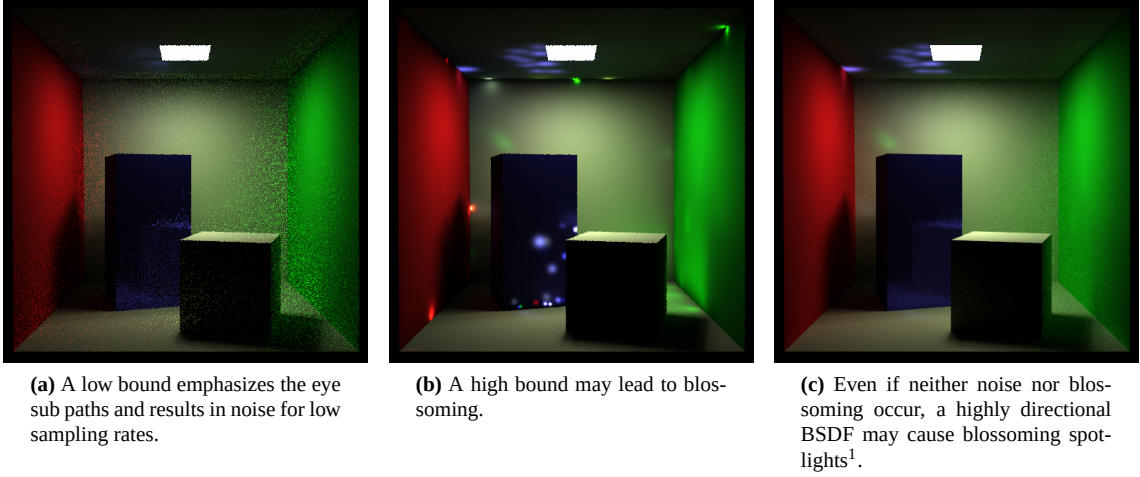


Figure 4.2: The choice of the bound and associated artifacts.

The optimal bound is not yet known but some ideas for a good bound have been developed by [Kollig and Keller, 2004] and Raab [2005]. Considering the BSDF at the eye subpath interaction point like in [Robust Singularity] is a good initial approach since a bound of the form $b = \frac{c}{f_d(\mathbf{x} - \mathbf{x} - \mathbf{x})}$ in combination with importance sampling according to the BSDF preserves the dominant features of scattering for the eye subpaths. The handling of specular surfaces is vastly improved compared to a constant bound, but the problems for light paths interacting with glossy and specular surfaces remain.

In addition to the shape of the bound the scaling factor c has to be determined. [Kollig and Keller, 2004] derived a scaling factor c for a single area light source by reasoning that the indirect lighting normally is

¹Although the bright spots caused by reusing light subpaths interaction points on a surface with a directional BSDF are similar to the blossoming due to the geometric term, we prefer to name the artifact *blossoming spotlight* in order to distinguish the different reasons for the artifact.

not brighter than direct lighting. Raab [2005] extended this to global illumination. Raab set the scaling factor c to the *potential direct irradiance* times a ratio γ . *Potential direct irradiance* denotes the irradiance that could be received directly from light sources, ignoring any visibility tests and directionality. The ratio γ is the ratio of potential direct irradiance to the potential direct irradiance arriving at a point. The value of gamma is guessed based on the reflectivity of the scene. A ratio of $\gamma = 3$ is considered a good choice for average scenes. The potential direct irradiance is related to the solid angle occupied by the light source when seen from the point of interest. Occlusion and orientation are ignored. The occupied solid angle is estimated by approximating the light source with an appropriately sized sphere. For multiple light sources the individual scaling factors are summed.

The complete illumination of a scene is the composition of the illumination by individual light sources. Thus it is legitimate to handle each light source individually to reduce the occurrence of improper values for the scale c . Each light subpath interaction point is part of a path that started at a light source. This light source provides the basic information for our bound. We approximate the solid angle for the calculation of the potential direct irradiance by multiplying the solid angle of the differential surface at the starting point of the current light subpath with the surface area of the light source. Occlusion and orientation are ignored.

4.1.1 Composite Bound

The blossoming spotlights artifact is more persistent than normal blossoming. We propose a new bound that reduces the problem of spotlighting artifacts. To account for the BSDF at the light subpath interaction point there are various possibilities. A practical approach is to adjust the shape of the near subdomain. The final choice should lessen the immediate contribution of interaction points with a focused BSDF if the principal direction of scattering coincides with the direction of the deterministic connection. Furthermore the near subdomain should be small and the calculations for the bound should be simple to keep the costs low. A smooth boundary of the near subdomain is preferred to avoid artifacts in the form of phantom contrasts.

The current bound is fine and accounts for the BSDF at the eye subpath interaction point, therefore we adopt it without change. The lobe defined by the inner product of the direction of the deterministic connection and an outgoing direction provides a smooth boundary for a subdomain. The attenuation of the deterministic connection by the BSDF at the light subpath interaction point is a good indicator for the blossoming spotlight artifact, therefore we use this attenuation value to scale the lobe. To avoid unnecessary inflation of the near subdomain we use the unification of the respective near subdomains. The formula for this bound is then given by

$$b = \frac{c}{\max \left\{ \widehat{\dot{\mathbf{x}}_e - \mathbf{x}_e \cdot \mathbf{x}_l - \mathbf{x}_e} f_s(\dot{\mathbf{x}}_l \rightarrow \mathbf{x}_l \rightarrow \mathbf{x}_e), f_s(\dot{\mathbf{x}}_e \rightarrow \mathbf{x}_e \rightarrow \dot{\mathbf{x}}_e) \right\}} \quad (4.3)$$

where the path has the form $(\dots, \dot{\mathbf{x}}_l, \mathbf{x}_l, \mathbf{x}_e, \dot{\mathbf{x}}_e, \dots)$ in case of the deterministic connection and $(\dots, \dot{\mathbf{x}}_e, \mathbf{x}_e, \dot{\mathbf{x}}_e, \dots)$ in case of the eye subpath extension.

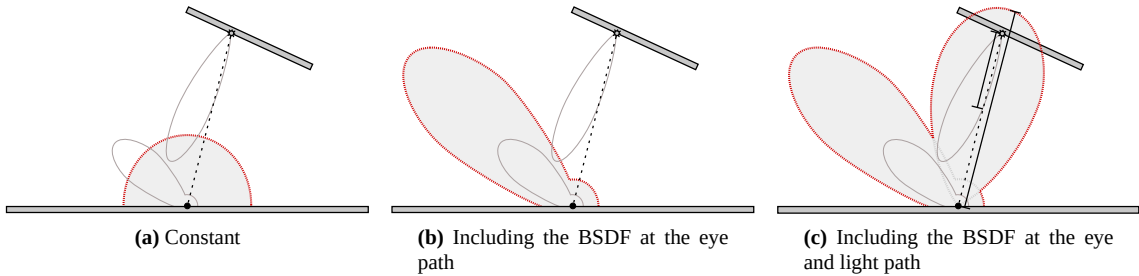


Figure 4.3: The near and far subdomains of the different bounds.

The problem of bright blossoming spotlight artifacts is reduced by the composite bound. The additional eye path extensions due to the larger near subdomain defined by the new bound are unsatisfactory at sampling caustic effects, but the resultant noise is much more tolerable than the former artifacts. A large near subdomain heavily hurts the overall efficiency of the RIGI algorithm due to the increased length of the eye

subpaths. Thus it seems advisable to use those bounds only in conjunction with the associated light subpath. This yields some overhead, because different bounds are used for different light subpaths, but is nevertheless practicable by accordingly adjusting the evaluation order of the subpath connections.

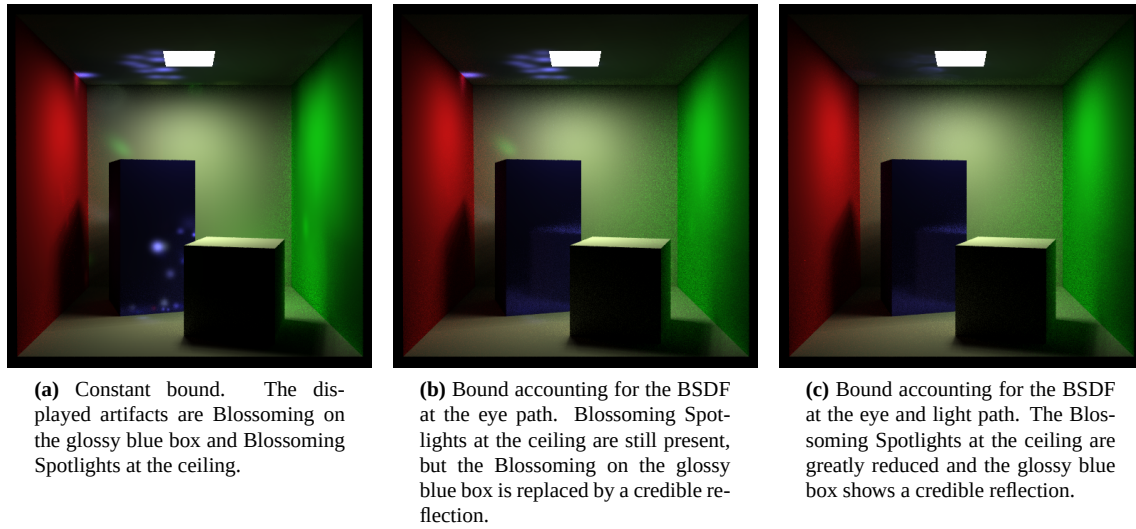


Figure 4.4: Comparison of images rendered with the different bounds.

A completely different approach to weighting strategies for RIGI is the extension of a classic multiple importance sampling heuristics. Unfortunately the classic heuristics based on the probability densities of all techniques have lower performance than the robust bounding heuristic as shown by Kollig and Keller [2004]. ctepkalosimp did some related work in his paper *Improved Indirect Photon Mapping with Weighted Importance Sampling*.

4.2 Backtraced Caustics

The sample density by light subpaths in RIGI normally does not suffice to adequately sample caustic effects. Large light sources allow for eye paths to sample caustics, too, but the smaller the light source, the smaller the chance to hit it. Instead of trying to capture caustic effects by using random eye subpaths, we aim eye subpath samples at light subpath interaction points, favoring those with a glossy BSDF.

To directly aim samples, we can either use the direction to the target point and perturb it or use the target position and perturb that. Both choices have their advantages: perturbation of the direction is easy to handle and normally provides a density in projected solid angle measure. Perturbation of the target point automatically adjusts the scale of the directional perturbation to the distance of the point, but requires transformations of the density. To primarily capture caustic effects, we would like to target primarily glossy and specular surfaces as proposed by [jensen?] for Photon Mapping. We use the reflection properties of the BSDF at the light subpath interaction points as a criterion for adding targeted samples.

The use of the light subpath interaction points as targets for sampling hints to an iterative solution to the problem of determining an optimal sample distribution for light transport. We further investigate this point in section 4.4.

Unfortunately the samples directly aimed at light subpath interaction points are not very successful at sampling caustics. Therefore we improve the position of the target points by estimating an reflection path. This equals targeting the reflection of the final target point on the light source. There are some analytic solutions, like that by [Mitchell and Hanrahan, 1992], for special object geometries, but for arbitrary scenes we have to rely on approximations. For our tests we use discrete differential information on the course of a path to extrapolate the mirror point of the target point. The technique of [Wand and Straßer, 2003] has some similarities to our approach.

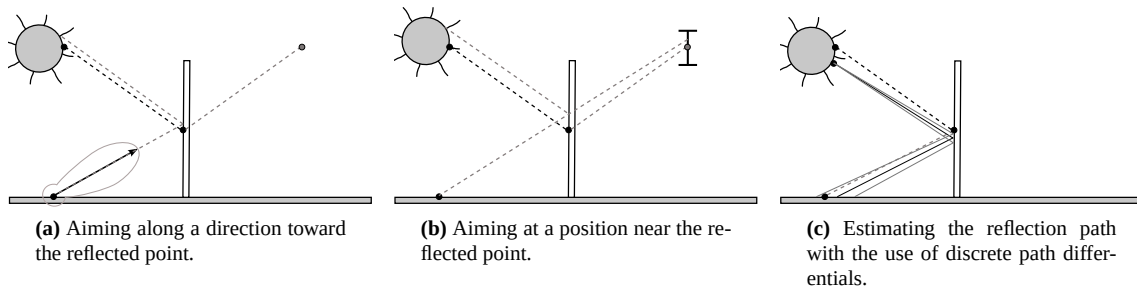


Figure 4.5: Different methods to backtrace caustics.

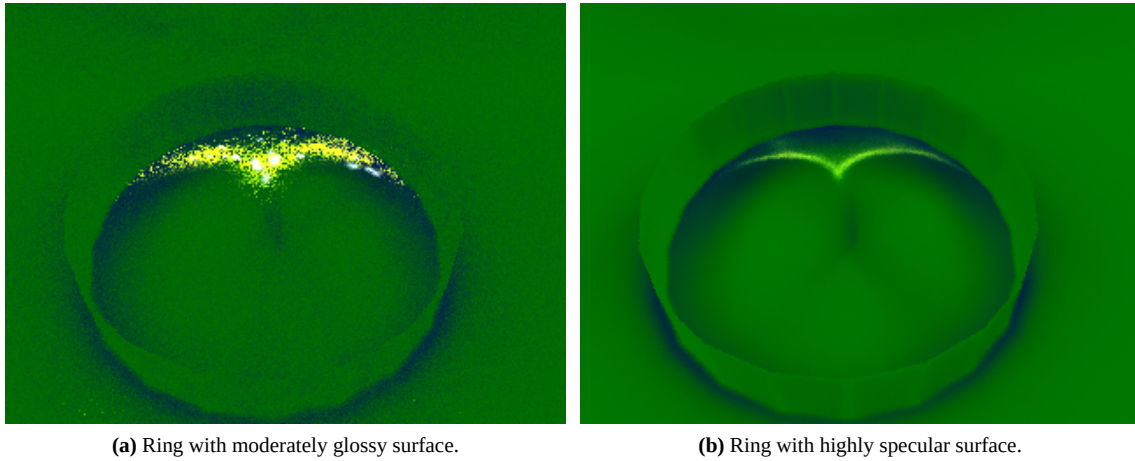


Figure 4.6: Low resolution false color images from a proof-of-concept implementation for backtracing caustics. The contribution of the backtraced light paths is colored yellow and the contribution of the reused light subpath interaction points is colored white.

We could produce satisfactory images for medium and large sized light sources but failed for small light sources. This subpaths can only be sampled efficiently by light subpaths. The proposed techniques remind of the *path differentials* introduced by [Igehy, 1999] and *Perturbation Methods for Interactive Specular reflections* by, which in our opinion have much potential to boost the efficiency of rendering algorithms.

4.3 Specifically Tailored Sampling Rates

As [Veach, 1997] has demonstrated and we have verified ourselves, BDPT captures directly seen caustics mainly by direct eye connections i.e. connections to eye subpaths with a single vertex. Something similar can be observed for caustics seen in a mirror. The previous attempt to use eye subpaths to sample those effects was only moderately successful, because this particular paths have a very low probability of being sampled by eye paths. Consequently we have to increase the light subpath sampling rate to sample these paths efficiently.

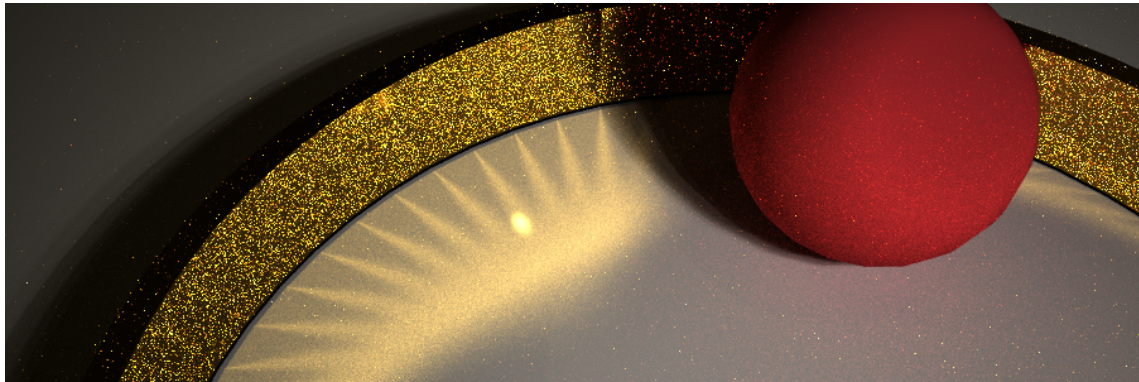
The sampling strategy of RIGI is tailored to capturing low frequency illumination effects efficiently. The additional samples necessary to capture high frequency illumination effects should not hamper the efficiency for low frequency illumination effects. For that purpose we have to separate the sampling techniques into two sets. One set is used to sample transport paths involved in low frequency illumination effects using a low sampling rate. The other set is used to sample transport paths involved in high frequency illumination effects applying an accordingly high sampling rate.

A possibility to achieve such a decomposition is to separate the techniques that apply a deterministic connection to two subpaths where the endpoint of the light subpath or its predecessor lies on a specular surface.

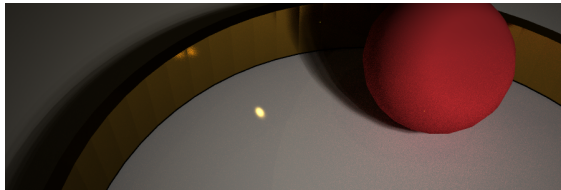
These techniques primarily sample the high frequency illumination effects entailed by specular surfaces and thus should receive an increased number of samples. We use the same eye paths for both sets of techniques and generate individual light subpaths for each eye subpath to achieve a high sampling rate. The respectively sampled paths for both sets of techniques are not disjoint, thus multiple importance sampling has to be applied to the combined sets of techniques. In fact, the multiple importance framework natively provides for the allocation of a different number of samples to different techniques. We retained the weighting by the robust bounding heuristic to avoid blossoming.

The criterion for a distinction of specular surfaces from non specular surfaces is arbitrary, but affects efficiency. In order to avoid artifacts, a continuous separation with appropriate weighting is preferred to a sharp binary separation of the techniques. The proposed third subdomain of the composite bound might be used to disjoin the paths sampled by the different sets of techniques.

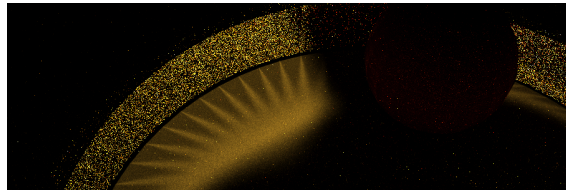
The separate sampling of high and low frequency illumination parallels the common use of disjoint photon maps for caustics and low frequency illumination proposed. In the original paper on RIGI it is actually suggested to apply photon mapping to separately capture high frequency illumination effects. The work on *Fast and Detailed Approximate Global Illumination by Irradiance Decomposition* by [Arikan et al., 2005] attempted a similar decomposition for illumination approximation.



(a) The combined image. The low frequency lighting is smooth as expected and the directly seen caustic is captured satisfactory. However, the mirror image of the caustic is sampled insufficiently. The paths necessary to capture such effects are extremely difficult to sample and require a much higher sampling rate than directly seen caustics.



(b) Contribution of the sparse samples.



(c) Contribution of the dense samples.

Figure 4.7: Images from rendering with individual sampling rates for the different sampling techniques for low and high frequency illumination effects. The scene is composed of a diffuse floor, a diffuse ball, a highly specular faceted ring, and a small area light source.

4.4 Efficient Point Light Sources

Instead of determining a decomposition of the sampling techniques to apply different sampling rates to different techniques a priori, we can attempt to adaptively adjust the sampling rate for light sub paths. While we can adjust to a single factor of the density of light transport by importance sampling according to the *BSDF*, we cannot easily account for the adjoint factor. We get a first estimate of the importance of the light path interaction points when a deterministic connection of eye paths and light paths is made, or a sensor is directly directly hit. The acquired product of importance and radiance at an interaction point equals the contribution of the complete path.

Instead of just adding more samples proportional to the contribution, we perform a *separation of the main part* and *importance sampling* on the remaining differences by using the technique presented in section 3.4.1. The calculated contribution serves as the main part. The variance of the contribution is a good indicator on the deviation of the final measurement from the current estimate.

In our proof-of-concept implementation of the algorithm we join every eye subpath with multiple pregenerated and stored light subpaths as usual. New light subpaths are added for each light subpath that results in a highly varying contribution when joined with different eye subpaths. For the additional paths we use the same random numbers as for the base path and perturb them before tracing the new path. This mutation strategy is originally the *Simple and Robust Mutation Strategy for the Metropolis Light Transport Algorithm* presented by [Kelemen et al., 2002]. The applied mutation does not change a underlying uniform distribution of samples since the mutation strategy is applied equally to every sample although some of the new paths are never used because Russian Roulette effected their omission. The mutation strategy increases the chance that the new samples are placed in the same regions of path space, where the base paths resulted in high variance. The adaptive sampling method from section 3.4.1 allows us to embed the supplementary light subpaths without introducing bias. The contributions from the individual paths are combined and stored per pixel to provide a memory efficient separable main part. The procedure can be repeated to progressively refine the sample distribution.

An increase of the sampling rate in conjunction with reuse is only practical if almost all samples carry valuable information, otherwise the work wasted on unprofitable samples will provoke inefficiency. The profit from different paths is erratic throughout the image. Therefore we expect a strongly increased performance of the proposed algorithm when applied to small parts of the image independently.

Chapter 5

Summary

During the preparation of this thesis much time has been spent on contemplating the available algorithms, their problems and on thinking of an general direction for expedient work on the topic of simulated light transport for image generation. This broad avenue entailed a large diversity of raised issues and a expanded section on potential future work.

We initially set out to reduce the *problem of many light sources*, especially in the context of the *Robust Instant Global Illumination Algorithm* by [Keller]. Contemplating the whole problem and related issues lead to a change of the primary objective. We concluded that our primary objective should be generic methods for the efficient sampling of path space instead of the reduction of visibility tests at one of the last steps of the rendering algorithm. The initial goal was an attempt to remedy a symptom. We chose to investigate the source of this symptom and to explore avenues to inhibit its occurrence.

First we developed a technique to efficiently calculate the direct illumination by many point light sources. We adapt the evaluation efforts to the predicted variance of the illumination. To keep the variance introduced by our technique low we rely on visibility and contribution approximation. Therefore we investigated the possibilities of visibility and contribution prediction.

In addition we conceived an unbiased variant of the Light Cuts algorithm by replacing the use of a deterministic error bound with our randomized technique. Furthermore the initially developed technique was generalized to cover the generic unbiased combination of approximations with corrective sampling. The technique was applied it to design an efficient and unbiased adaptive sampling framework.

The Robust Instant Global Illumination algorithm was examined to reveal its weak spots. Based on the obtained insight we proposed some extensions to the algorithm to increase its range of application. We remedied the problem of blossoming spotlights by a composite bound for use with the robust bounding heuristic to permit the application of the algorithm to scenes with glossy surfaces. Next we attempted to backtrace the light subpaths involved in the creation of caustic effects, but had limited success. Thus we took the other route and directly increased the sampling rate for that part of the light subpaths and obtained a remarkably effective rendering algorithm adaptable to high and low frequency illumination effects. Finally we presented a method to generate efficient point light sources adapted to the characteristics of light transport.

The principal achievement of this thesis is not a particular highly optimized algorithm with a detailed assessment, but a batch of - more or less - novel approaches to - more or less - general problems, encountered when working on the simulation of light transport for image generation.

5.1 Future Work and Extensions

It is important to note that the presented ideas for future work have not yet been checked for coincidences with existing solutions. Some of the proposals may have been already implemented.

Refined the Preprocess: A very influential part of RIGI is the preprocess of generating and storing light subpaths. The generated and stored light subpaths are extensively reused for the whole image and thus have a large impact on the quality. Many techniques originally developed for improved Photon Map generation may be adapted. Promising candidates are the *Efficient Importance Sampling Techniques for the Photon Map* by [Keller and Wald, 2000].

Go with the Winners for Light Subpaths: A efficiency gain is expected from application of the *Go with the Winners Strategy* by [Szirmay-Kalos et al., 2005] to the preprocess. This strategy introduces splitting to the generation of the light subpaths. The number of splits is adapted to the predicted variance of the contribution of the path, thus increasing the efficiency of the light subpaths.

Multiple Sets of Light Subpaths for Robust Instant Global Illumination: Most of the reused light subpaths are accountable for a very different level of contribution in different regions of the image, therefore different sets of light subpaths are necessary for different image areas. The extent of a distinct image area can be determined by tiling, interleaving, or any other method. There are no special requirements on the shape of the tiles since it is unproblematic to combine multiple overlapping images to a single image. Irregular tiles are perfectly valid. To *Accelerate Path Tracing by Re-Using Paths*, [Bekaert et al., 2002] uses N-rooks like patterns as tiles to replace the blocky artifacts of regular tiles by structured noise. Even the fragmented subimages used for interleaved sampling by [Keller and Heidrich, 2001] can be interpreted as a kind of tiles. It is sensible to gradually change the contents of the used set while traversing the image to avoid artifacts from switching the used set of light subpaths. A simplistic implementation for these *floating sets* could proceed as follows: First references to all light subpaths are arranged in an array. To determine a set of light subpaths for each pixel position, a window is floated along the rows and columns of the array paralleling the change of the pixel position. All light subpaths that are referenced inside the window are part of the local set.

Information Reuse by Grouping: An extremely simple form of reuse is the use of enlarged path families for full BDPT. Instead of consecutively combining a single eye subpath and a single light subpath to form a family of complete paths, the path family can be extended to all combinations of multiple eye subpaths with multiple light subpaths. This approach reuses each eye subpath for combination with a number of light subpaths and vice versa. To avoid artifacts, the content of the groups should be replaced gradually. Similar attempts could be made to reintroduce reuse of transport paths to the *Robust Metropolis Light Transport* algorithm discussed by [Seibert, 2006].

Reintroduction of Reconstruction Cuts to Multidimensional Light Cuts: *Light Cuts* is an algorithm that applies reuse on the base of hierarchical light source cluster. The extension *Multidimensional Light Cuts* by [Walter et al., 2006] additionally clusters the eye path interaction points per pixel. We believe the cluster building for light subpaths not necessarily has to stop at pixel level, but can be extended to tiling to reintroduce the *reconstruction cuts* of the original *Light Cuts* algorithm.

Appendix A

Derivation of the Efficiency of Unbiased Adaptive Sampling

Since the efficiency of an estimator is defined as $\epsilon = \frac{1}{VT}$ where V is the variance of the estimator, we have to calculate the variance of the estimator $\tilde{F}_n$.

$$E[\tilde{F}_n] = E\left[\frac{n-1}{n}\tilde{F}_{n-1} + \frac{1}{n}\left(\tilde{F}_{n-1} + \begin{cases} \frac{1}{p}(\tilde{e}(f) - \tilde{F}_{n-1}) & \text{for } \xi \leq p \\ 0 & \text{otherwise} \end{cases}\right)\right] \quad (.1a)$$

$$= \frac{n-1}{n}E[\tilde{F}_{n-1}] + \frac{1}{n}E[\tilde{e}(f)] \quad (.1b)$$

$$E[\tilde{F}_n]^2 = \left(\frac{n-1}{n}E[\tilde{F}_{n-1}] + \frac{1}{n}E[\tilde{e}(f)]\right)^2 \quad (.2a)$$

$$= \frac{1}{n^2} \left((n^2 - 2n + 1)E[\tilde{F}_{n-1}]^2 + 2(n-1)E[\tilde{F}_{n-1}]E[\tilde{e}(f)] + E[\tilde{e}(f)]^2 \right) \quad (.2b)$$

$$E[(\tilde{F}_n)^2] = E\left[\left(\tilde{F}_{n-1} + \frac{1}{n}\begin{cases} \frac{1}{p}(\tilde{e}(f) - \tilde{F}_{n-1}) & \text{for } \xi \leq p \\ 0 & \text{otherwise} \end{cases}\right)^2\right] \quad (.3a)$$

$$= E\left[\begin{cases} \tilde{F}_{n-1}^2 + 2\tilde{F}_{n-1}\frac{1}{pn}(\tilde{e}(f) - \tilde{F}_{n-1}) + \frac{1}{p^2n^2}(\tilde{e}(f) - \tilde{F}_{n-1})^2 & \text{for } \xi \leq p \\ \tilde{F}_{n-1}^2 & \text{otherwise} \end{cases}\right] \quad (.3b)$$

$$= E[\tilde{F}_{n-1}^2] + E\left[2\frac{1}{n}(\tilde{e}(f)\tilde{F}_{n-1} - \tilde{F}_{n-1}^2) + \frac{1}{pn^2}(\tilde{e}(f)^2 - 2\tilde{F}_{n-1}\tilde{e}(f) + \tilde{F}_{n-1}^2)\right] \quad (.3c)$$

$$= (1 - \frac{2}{n} + \frac{1}{pn^2})E[\tilde{F}_{n-1}^2] + (\frac{2}{n} - \frac{2}{pn^2})E[(\tilde{e}(f)\tilde{F}_{n-1})] + \frac{1}{pn^2}E[\tilde{e}(f)^2] \quad (.3d)$$

$$= \frac{1}{n^2} \left((n^2 - 2n + \frac{1}{p})E[\tilde{F}_{n-1}^2] + 2(n - \frac{1}{p})E[(\tilde{e}(f)\tilde{F}_{n-1})] + \frac{1}{p}E[\tilde{e}(f)^2] \right) \quad (.3e)$$

$$= \frac{1}{n^2} \left((n^2 - 2n + \frac{1}{p})(E[\tilde{F}_{n-1}^2] - E[\tilde{F}_{n-1}]^2) + E[\tilde{F}_{n-1}]^2 \right) \quad (.3f)$$

$$+ 2(n - \frac{1}{p})E[(\tilde{e}(f)\tilde{F}_{n-1})] + \frac{1}{p}(E[\tilde{e}(f)^2] - E[\tilde{e}(f)]^2 + E[\tilde{e}(f)]^2) \quad (.3g)$$

$$= \frac{1}{n^2} \left((n^2 - 2n + \frac{1}{p})V[\tilde{F}_{n-1}] + \frac{1}{p}(V[\tilde{e}(f)]) \right) \quad (.3h)$$

$$+ \frac{1}{n^2} \left((n^2 - 2n + \frac{1}{p})E[\tilde{F}_{n-1}]^2 + 2(n - \frac{1}{p})E[(\tilde{e}(f)\tilde{F}_{n-1})] + \frac{1}{p}E[\tilde{e}(f)]^2 \right) \quad (.3i)$$

If $\tilde{F}_{n-1}$ and $\tilde{e}(f)$ are independent and thus $E[\tilde{F}_{n-1}\tilde{e}(f)] = E[\tilde{F}_{n-1}]E[\tilde{e}(f)]$ we can calculate the variance

of F_n by

$$V[\tilde{F}_n] = E[(\tilde{F}_n)^2] - E[\tilde{F}_n]^2 \quad (.4a)$$

$$= \frac{1}{n^2} \left((n^2 - 2n + \frac{1}{p})V[\tilde{F}_{n-1}] + \frac{1}{p}(V[\tilde{e}(f)]) \right) \quad (.4b)$$

$$+ \frac{1}{n^2} \left((n^2 - 2n + \frac{1}{p})E[\tilde{F}_{n-1}^2] + 2(n - \frac{1}{p})E[\tilde{e}(f)\tilde{F}_{n-1}] + \frac{1}{p}E[\tilde{e}(f)^2] \right) \quad (.4c)$$

$$- \frac{1}{n^2} \left((n^2 - 2n + 1)E[\tilde{F}_{n-1}]^2 + 2(n - 1)E[\tilde{F}_{n-1}]E[\tilde{e}(f)] + E[\tilde{e}(f)]^2 \right) \quad (.4d)$$

$$= \frac{1}{n^2} \left((n^2 - 2n + \frac{1}{p})V[\tilde{F}_{n-1}] + \frac{1}{p}V[\tilde{e}(f)] \right) + \frac{1}{n^2}(\frac{1}{p} - 1)(E[\tilde{e}(f)] - E[\tilde{F}_{n-1}])^2. \quad (.4e)$$

The term $(E[\tilde{e}(f)] - E[\tilde{F}_{n-1}])^2 = 0$ can be omitted if the expected values match. This leaves us with:

$$V[\tilde{F}_n] = V[\tilde{F}_{n-1}] + \frac{1}{n^2} \left((\frac{1}{p} - 2n)V[\tilde{F}_{n-1}] + \frac{1}{p}V[\tilde{e}(f)] \right). \quad (.5)$$

An extension of the estimator $\tilde{F}_{n-1}$ to the estimator $\tilde{F}_n$ reduces the variance if

$$0 \geq \frac{1}{n^2} \left((\frac{1}{p} - 2n)V[\tilde{F}_{n-1}] + \frac{1}{p}V[\tilde{e}(f)] \right) \quad (.6a)$$

$$p \cdot 2V[\tilde{F}_{n-1}] \geq \frac{1}{N} \left(V[\tilde{F}_{n-1}] + V[\tilde{e}(f)] \right) \quad (.6b)$$

$$p \geq \frac{V[\tilde{F}_{n-1}] + V[\tilde{e}(f)]}{2nV[\tilde{F}_{n-1}]} \quad (.6c)$$

$$p \geq \frac{1 + \frac{V[\tilde{e}(f)]}{V[\tilde{F}_{n-1}]}}{2n}. \quad (.6d)$$

The efficiency of an estimator $\tilde{F}_{n-1}$ with a variance of $V(\tilde{F}_{n-1})$ and a cost of $T(\tilde{F}_{n-1})$ is

$$\epsilon_{\tilde{F}_{n-1}} = \frac{1}{V(\tilde{F}_{n-1})T(\tilde{F}_{n-1})} \quad (.7)$$

The efficiency of the extended estimator $\tilde{F}_n$ is

$$\epsilon_{\tilde{F}_n} = \frac{1}{\left(V[\tilde{F}_{n-1}] + \frac{1}{n^2} \left((\frac{1}{p} - 2n)V[\tilde{F}_{n-1}] + \frac{1}{p}V[\tilde{e}(f)] \right) \right) \left(T(\tilde{F}_{n-1}) + T(\tilde{e}) \right)}. \quad (.8)$$

The efficiency of the extended estimator $\tilde{F}_n$ is higher than the efficiency of its the unextend estimator $\tilde{F}_{n-1}$

if

$$\epsilon_{\tilde{F}_{n-1}} \leq \epsilon_{\tilde{F}_n} \quad (.9a)$$

$$V(\tilde{F}_{n-1})T(\tilde{F}_{n-1}) \geq \left(V[\tilde{F}_{n-1}] + \frac{1}{n^2} \left(\left(\frac{1}{p} - 2n \right) V[\tilde{F}_{n-1}] + \frac{1}{p} V[\tilde{e}(f)] \right) \right) \left(T(\tilde{F}_{n-1}) + T(\tilde{e}) \right) \quad (.9b)$$

$$0 \geq V(\tilde{F}_{n-1})T(\tilde{e}) + \frac{1}{n^2} \left(\left(\frac{1}{p} - 2n \right) V(\tilde{F}_{n-1}) \left(T(\tilde{F}_{n-1}) + T(\tilde{e}) \right) + \frac{1}{p} V(\tilde{e}) \left(T(\tilde{F}_{n-1}) + T(\tilde{e}) \right) \right) \quad (.9c)$$

$$0 \geq V(\tilde{F}_{n-1})T(\tilde{e}) - \frac{2}{n} V(\tilde{F}_{n-1}) \left(T(\tilde{F}_{n-1}) + T(\tilde{e}) \right) + \frac{1}{n^2} \frac{1}{p} \left(\left(V(\tilde{F}_{n-1}) + V(\tilde{e}) \right) \left(T(\tilde{F}_{n-1}) + T(\tilde{e}) \right) \right) \quad (.9d)$$

$$2NV(\tilde{F}_{n-1}) \left(T(\tilde{F}_{n-1}) + T(\tilde{e}) \right) - n^2 V(\tilde{F}_{n-1})T(\tilde{e}) \geq \frac{1}{p} \left(\left(V(\tilde{F}_{n-1}) + V(\tilde{e}) \right) \left(T(\tilde{F}_{n-1}) + T(\tilde{e}) \right) \right) \quad (.9e)$$

$$p \geq \frac{\left(V(\tilde{F}_{n-1}) + V(\tilde{e}) \right) \left(T(\tilde{F}_{n-1}) + T(\tilde{e}) \right)}{2nV(\tilde{F}_{n-1}) \left(T(\tilde{F}_{n-1}) + T(\tilde{e}) \right) - n^2 V(\tilde{F}_{n-1})T(\tilde{e})}. \quad (.9f)$$

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Erklärung

Name: Manuel Jan Kugelman

Matrikel-Nr.: 468340

Ich erkläre, dass ich die Diplomarbeit selbstständig verfasst und keine
anderen als die angegebenen Hilfsmittel verwendet habe.

Ulm, den